

Structure-Guided Reinforcement Learning for Partially Observed Multi-Period Portfolio Optimization with Transaction Costs

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Abstract

This paper develops a structure-guided learning framework for multi-period portfolio optimization with latent market regimes and proportional transaction costs. Stage 1 studies a precommitted mean–variance problem. Under common observation covariance, Wonham filtering yields a belief-state diffusion, while the entropy-regularized HJB equation identifies a Gaussian exploratory policy and a wealth-quadratic value function. These results motivate a low-dimensional actor–critic rather than an unrestricted state-to-weight learner. A statistical jump model supplies regime-dependent parameter estimates to the online filter. For regime-dependent covariance, an exact Gaussian Bayes correction for the adopted discrete-time hidden Markov observation model is combined with a moment-matched control approximation and an approximate HJB structure.

Stage 2 freezes the frictionless policy as a state-dependent center for costly trading. A transaction-cost QVI provides the theoretical benchmark, but joint critic–intervention learning was numerically unstable; the reported implementation therefore learns asymmetric no-trade boundaries through direct pathwise optimization.

Across five independent training seeds evaluated on a common set of 1000 held-out paths, the common-covariance Stage 1 policy was highly stable, with mean terminal wealth 1.1811 and a 95% Student- t interval across training seeds of $[1.1808, 1.1814]$. Center+DNNBand achieved mean net terminal wealth 1.2009 with interval $[1.1807, 1.2211]$, exceeded daily center rebalancing by 0.0910 with interval $[0.0709, 0.1112]$, and reduced mean trade count from 252 to 175.8. Its descriptive difference from Stage 1 was 0.0198 with interval $[-0.0003, 0.0400]$; because this interval includes zero and the two stages optimize different objectives, no Stage 1-versus-Stage 2 dominance claim is made. The regime-dependent-covariance experiment remains exploratory and single-seed. The framework is interpreted as a portfolio-control building block for institutional and actuarial applications rather than a complete asset–liability management model.

Keywords: continuous-time reinforcement learning; mean–variance portfolio selection; partial information; regime switching; transaction costs; no-trade region; actor–critic.

1 Introduction

Dynamic portfolio choice becomes substantially harder when three realistic features are introduced simultaneously: investment opportunities vary across persistent market regimes, the regime is not directly observed, and trading incurs transaction costs. Partial information requires the investor to infer the state of the market from prices. Multi-period objectives require decisions to account for their effect on future wealth and information. Transaction costs make frequent rebalancing suboptimal and typically replace frictionless point targets by state-dependent no-trade regions. These features lead naturally to stochastic filtering, nonlinear dynamic programming, and high-dimensional policy optimization.

The starting point of this study is the natural compatibility between portfolio choice and reinforcement learning. A portfolio rule is a policy that maps an information state into an asset allocation, while reinforcement learning is designed to learn state-dependent sequential decisions from interaction data. The central issue is therefore not only which learning algorithm to use, but also how to construct the state, policy class, and objective. Here financial domain knowledge enters all three components: filtering converts return histories into an economically interpretable regime belief; stochastic-control theory restricts the actor and critic to HJB-consistent forms; and the objective incorporates target wealth, exploration, turnover, and transaction costs. This structure is intended to improve sample efficiency, numerical stability, and interpretability relative to an unrestricted state-to-weight learner. The five-seed common-covariance experiments assess robustness to training initialization, but they do not isolate the contribution of each structural restriction through a controlled structured-versus-unstructured ablation; repeated-seed robustness and ablation are distinct open questions.

This compatibility does not imply that reinforcement learning is necessary for every low-dimensional portfolio problem. Its role here is computational: it provides a simulation-based policy-approximation method when

partial information, nonlinear objectives, and transaction costs make direct dynamic programming difficult to implement.

This restriction is particularly important in regime-switching markets. Abrupt changes, persistent latent states, noisy regime estimates, and delayed terminal rewards can make generic deep-RL policies difficult to train. Preliminary experiments with less structured deep architectures, including Transformer-style state encoders and an unrestricted transaction-cost policy, did not yield a sufficiently reliable policy for the reported study. These trials do not isolate a single cause: policy-class flexibility, terminal credit assignment, state normalization, pathwise-gradient variance, hyperparameters, and training budget may all contribute. Constraining the output to economically meaningful, center-relative policy geometry provided a stronger inductive bias. The project therefore uses financial mathematics to determine the belief state and feasible policy geometry, while leaving only lower-dimensional coefficients or boundaries to be learned. For the costly problem, the frictionless solution is used as an advanced start so that Stage 2 learns a correction around an economically meaningful benchmark rather than the full policy from scratch.

Regime changes are also not limited to expected returns. Volatility levels and cross-asset correlations can change materially across market conditions, and these covariance shifts directly affect diversification, risk budgets, capital requirements, and the scale of admissible positions. For practical portfolio and risk management, covariance misspecification may therefore be at least as consequential as drift misspecification. This motivates the regime-dependent-covariance extension, while the common-covariance model is retained as the theorem-level benchmark needed to identify which HJB structures can be transferred safely.

This paper combines two complementary components. Stage 1 learns a frictionless portfolio policy under partial information. It starts from the continuous-time precommitted mean–variance formulation of Wu and Li [16], extends its structured actor–critic representation to multiple risky assets, and makes the filtering, policy, value function, and learning pipeline operational. Stage 2 first formulates the transaction-cost problem as a QVI on the enlarged state (t, x, p, w) . Because the corresponding critic-based RL implementation was not numerically stable in the current experiments, the reported method uses Center+DNNBand, the implemented Direct Boundary RL policy: a neural policy predicts lower and upper no-trade boundaries around the frozen Stage 1 center and is trained by pathwise differentiation through the wealth recursion with proportional costs. We refer to this evaluated policy as Center+DNNBand throughout the remainder of the paper. The reported Stage 2 does not claim to solve the exact transaction-cost extension of the Stage 1 mean–variance problem: it uses the Stage 1 policy as a state-dependent structural feature and optimizes a separate transaction-cost-aware terminal objective.

The organizing principle is *structure-guided learning*. In Stage 1, the observable state is determined by filtering, the exploratory policy family is determined by the HJB, and the critic is restricted by the quadratic terminal condition. In Stage 2, classical transaction-cost theory motivates an inaction region, while the Stage 1 policy supplies a state-dependent anchor. Learning is therefore used to estimate a small set of economically meaningful objects rather than an unrestricted state-to-action map.

Across the contributions, the common methodological principle is selective function approximation: low-dimensional parametrization is used where the control equation identifies the policy form, while neural approximation is reserved for residual boundary geometry that remains analytically unresolved. The distinction between the two stages is therefore not simply parametric learning versus deep learning; it reflects how sharply the underlying control theory identifies the admissible functional form.

The paper makes the following four contributions.

- (i) **SJM-based regime estimation for regime-switching GBM control.** We integrate the statistical jump model idea into partially observed multi-asset portfolio optimization under a regime-switching geometric Brownian motion. Persistent offline SJM labels are converted into regime-specific drift, covariance, and transition-intensity estimates, which are then supplied to the online Bayesian filter. The contribution is the integration of an established regime-identification method with the HJB-guided control pipeline, rather than a new statistical theory for SJM.
- (ii) **An approximate HJB extension to regime-dependent covariance.** To retain an HJB-guided policy structure when volatility and correlation vary by regime, we combine an exact Gaussian Bayes correction for the specified discretized observation model with a moment-matched, belief-averaged state approximation. We derive the resulting approximate HJB, Gaussian exploratory policy, and coefficient equations, including the nonlinear term that vanishes in the common-covariance benchmark. This is an approximate theoretical contribution; no verification or approximation-error theorem is claimed. In a separate exploratory single-seed experiment, the surrogate produced higher mean terminal wealth and lower dispersion than the displayed static baselines. This result is not part of the five-seed Phase 1A robustness analysis, and its noisier training indicates that convergence and approximation quality remain open issues.

- (iii) **A two-stage learning architecture for portfolio optimization with transaction costs.** Stage 1 learns a frictionless HJB-guided policy, which is then frozen as a state-dependent center for Stage 2. We formulate the costly problem through a QVI and derive a center-relative critic/intervention parametrization consistent with its continuation and intervention structure. Because joint QVI-residual and intervention learning was numerically unstable, the reported implementation retains the robust structural implications—a frictionless center, an inaction region, feasibility constraints, and pathwise transaction costs—and learns asymmetric boundaries directly. The contribution is a structured two-stage framework that separates frictionless policy learning from transaction-cost correction while optimizing a separate transaction-cost-aware terminal objective in Stage 2. Across five training seeds, the implemented boundary policy retained a positive paired net-wealth difference relative to the daily and monthly center-rebalancing comparators, while materially reducing trade frequency relative to daily rebalancing.
- (iv) **A TD-inspired implementation refinement for continuous-time actor-critic learning.** The Stage 1 actor update combines a local martingale increment, which plays a temporal-difference-like role, with a normalized full-horizon tail residual, which provides a Monte Carlo-like terminal correction. Together with the critic update, this mixed signal is designed to address finite-horizon credit assignment and provided a usable implementation in the reported experiments. It is presented as an algorithmic refinement rather than a new continuous-time RL theorem; a controlled actor-signal ablation remains to be completed.

The common-covariance Stage 1 and Stage 2 experiments above are empirically validated on a common set of independently generated evaluation paths, repeated across five independent training seeds. For each training seed, the same fixed evaluation scenarios are used to compute seed-level summary statistics; these are then combined across seeds into a mean, a between-seed standard deviation, and a 95% Student- t interval with four degrees of freedom. Where two policies are compared on the same paths, paired differences are computed at the evaluation-episode level before aggregation across seeds. This design quantifies sensitivity to training initialization under a fixed market-evaluation design; it is not a claim about market-distribution uncertainty. Under this protocol, Stage 2 shows materially greater training-seed variability than Stage 1.

Actuarial relevance and potential applications

The framework is motivated by long-horizon institutional investment problems faced by insurers, pension funds, and other liability-aware investors. In such settings, the economically relevant decision is not only the current asset allocation but also the future sequence of rebalancing decisions. Market conditions are persistent but not directly observable, parameter estimates are noisy, and large investors face bid-ask spreads, commissions, market impact, operational constraints, and governance limits on trading frequency. These features make partial information and transaction costs central rather than secondary modeling details.

The target terminal wealth z can be interpreted as a reduced-form funding or accumulation target related to future benefit payments, a required funding ratio, a credited-rate objective, or a long-term reserve target. The belief p_t provides an explicit summary of uncertainty about the prevailing market environment, while a no-trade region translates the trade-off between tracking a desired allocation and avoiding unnecessary turnover into an explainable decision rule. This is potentially useful in settings where investment policies must be communicated to risk committees and implemented subject to transaction and governance budgets.

The present model is not a full asset-liability management model. It does not include stochastic liabilities, solvency capital, regulatory constraints, cash-flow matching, or liability-dependent utility. Instead, it provides a portfolio-control building block that could be embedded in a broader ALM framework by replacing the terminal target and admissible set with liability- and capital-sensitive specifications.

Table 1 summarizes how the main components are positioned relative to prior literature. The table supplements, rather than replaces, the contribution statements above.

Table 1: Positioning of the paper relative to prior literature.

Component	Prior literature	This paper
Stage 1 control	One-risky-asset hidden-regime benchmark; common-covariance multi-asset extension	Operational multi-asset actor-critic with explicit coefficient PDEs and implementation details
Regime estimation	SJM-based persistent regime identification	SJM labels integrated into parameter estimation for online belief-state control
Covariance structure	Common observation covariance in the theorem-level benchmark	Discretized Markov prediction and exact Gaussian Bayes correction plus an approximate belief-averaged HJB control layer
Transaction costs	Classical no-trade regions and HJB-initialized boundary networks	Belief-conditioned two-stage learning with a frozen Stage 1 center and Center+DNNBand
Actor update	Martingale-based continuous-time policy evaluation and gradient methods	Mixed local martingale and full-horizon rollout signal for finite-sample learning

2 Related Literature

2.1 Continuous-time reinforcement learning and mean-variance control

The continuous-time reinforcement-learning foundation used here follows the exploratory stochastic-control formulation of Wang, Zariphopoulou, and Zhou [15]. Randomized controls and entropy regularization convert exploration into part of the control objective. In linear-quadratic problems, this construction naturally produces Gaussian policies. Wang and Zhou [14] apply the framework to continuous-time mean-variance portfolio selection and show how analytical structure can substantially reduce the number of learned parameters.

Jia and Zhou [4] develop a martingale characterization of continuous-time policy evaluation. A central insight is that a naive mean-square temporal-difference error converges toward quadratic variation and is not, by itself, the appropriate continuous-time value-fitting objective. They instead derive martingale-loss and martingale-orthogonality formulations that recover continuous-time analogues of Monte Carlo and TD methods. Their companion paper [5] derives policy-gradient representations and offline and online actor-critic algorithms. The policy-gradient problem is recast through policy evaluation, while online updates use martingale orthogonality and stochastic approximation. These results motivate the martingale-based critic and actor signals used in Stage 1.

2.2 Partial information and regime-switching portfolios

Partial-information portfolio problems separate state estimation from control when an appropriate filtering reduction is available. Wonham [17] derives the finite-state nonlinear filter used in the common-covariance benchmark, while Krishnamurthy [7] develops the broader POMDP perspective in which the posterior distribution becomes the fully observed belief state. Xiong and Zhou [18] establish a separation principle for continuous-time mean-variance selection under partial information, and Elliott, Siu, and Badescu [3] study estimation and filtering under hidden Markov regimes.

For observable market regimes, Zhou and Yin [19] derive continuous-time mean-variance results and efficient-frontier implications. Under unobservable Markov-modulated drift, Rieder and Bäuerle [10] study portfolio optimization after filtering and identify the interaction between current opportunities and hedging against belief changes. These papers clarify the role of regime information and motivate the myopic-hedging decomposition used in the present Stage 1 actor.

The closest benchmark is Wu and Li [16]. They formulate an entropy-regularized, precommitted mean-variance problem with an unobservable market regime, use Wonham filtering to obtain a belief-state problem, and derive semi-analytical representations of the value function and Gaussian optimal policy. The unknown coefficient functions solve lower-dimensional parabolic PDEs and are approximated by polynomials inside an actor-critic algorithm. The present Stage 1 follows this logic, makes the multi-asset extension explicit, and adds a practical regime-estimation pipeline and a regime-dependent covariance experiment.

2.3 Statistical jump models for regime identification

Statistical jump models estimate persistent market regimes by clustering temporally constructed features while directly penalizing changes in the latent label sequence. This formulation offers a non-likelihood-based alternative to a conventional hidden Markov model and is particularly attractive when financial regimes are persistent, imbalanced, or supported by limited data. Aydinhan, Kolm, Mulvey, and Shu [1] review the discrete statistical jump model and extend its discrete hidden state to a probability vector on the regime simplex. Their continuous statistical jump model introduces smooth regime probabilities, derives a probabilistic interpretation of the transition penalty, and adds a mode-loss penalty that encourages decisive regime assignments. Their simulations and market experiments report improved robustness relative to discrete jump models and standard HMM estimation, especially under regime imbalance and short samples.

The present paper uses the simpler discrete two-state SJM rather than the continuous extension of Aydinhan et al. Its role is also narrower: the SJM provides a persistent offline segmentation from which regime-specific drifts, covariances, and empirical transition rates are estimated. It does not replace the online Wonham or Gaussian Bayes filter used to construct the belief state for control. Thus, the contribution here is the integration of SJM-based estimation with an HJB-guided portfolio-learning architecture, not a new SJM estimator or a new statistical consistency result.

The choice of SJM does not imply that it uniformly dominates hidden Markov models. HMMs provide a coherent probabilistic observation model and naturally generate filtered and smoothed regime probabilities. The discrete SJM used here instead offers direct control over label persistence through the jump penalty and produces regime prototypes in an economically constructed feature space. This makes the persistence penalty, feature centroids, and resulting offline segmentation particularly transparent when the objective is to identify persistent regimes rather than to estimate a complete generative return model.

The two approaches play complementary roles in the present pipeline. SJM is used offline to segment historical or simulated observations and estimate regime-dependent moments and transition intensities. Online allocation does not treat the resulting hard SJM label as an observed state; it uses a Bayesian filter to update the posterior regime probability from newly observed returns. This separation avoids supplying an ex-post label directly to the controller and preserves the partial-information structure of the investment problem.

2.4 Transaction costs and no-trade regions

The no-trade-region principle originates in classical proportional-cost portfolio theory. Magill and Constantinides [8] show that investors optimally tolerate deviations from frictionless target proportions. Davis and Norman [2] provide a rigorous infinite-horizon formulation and numerical characterization, and Shreve and Soner [11] establish existence, uniqueness, and regularity results using viscosity-solution methods. Taksar, Klass, and Assaf [13] provide a diffusion model with brokerage fees in which the risky-to-safe wealth ratio is kept inside an interval with the minimum intervention needed at its boundaries. Together, these studies supply the economic and QVI intuition for Stage 2: proportional costs create an inaction region rather than a continuously tracked point target.

High-dimensional PDE methods motivate neural approximations when grid-based QVI solvers become impractical. Han, Jentzen, and E [6] and Sirignano and Spiliopoulos [12] develop deep-learning methods for high-dimensional PDEs. In portfolio optimization, Mulvey et al. [9] solve a frictionless HJB system for mean-reverting Ornstein–Uhlenbeck assets to obtain an advanced start and then use a feedforward neural network to approximate state- and time-dependent no-trade boundaries. Their experiments demonstrate scalability to high-dimensional collections of correlated pair trades. The present work adopts the same broad principle—combine a tractable frictionless solution with a learned transaction-cost correction—but differs in three respects. The underlying market is regime-switching rather than OU, the state contains a filtered belief, and the reported Stage 2 network directly optimizes a separate terminal utility and regularization objective through simulated wealth paths. Accordingly, the Stage 1 center is a structural feature rather than the exact zero-cost optimizer of the Stage 2 objective.

3 Decision Framework and Partially Observed Market

3.1 From partial observation to belief-state control

A reinforcement-learning problem is described by an information state, an action, a policy, and an objective. In this portfolio application, the action is an allocation of risky exposure, while the information state must summarize what is relevant for both future wealth dynamics and future decisions. In Stage 1 the observable

control state is (t, X_t, p_t) , where X_t is discounted wealth and p_t is the posterior probability of regime 1. In Stage 2 the current holdings or portfolio weights are added because transaction costs depend on the distance between pre-trade and post-trade positions.

The underlying regime I_t is not observed directly. The investor observes the price or return history $\mathcal{F}_t^S$, so the original problem is a partially observed Markov decision process. Under the filtering assumptions used below, the posterior distribution of the finite-state regime is a sufficient statistic for that observation history. The problem can therefore be reformulated as a fully observed Markov control problem on the *belief state* [7]. This transformation does not eliminate regime uncertainty: it represents that uncertainty explicitly through

$$p_t = \mathbb{P}(I_t = 1 \mid \mathcal{F}_t^S). \quad (1)$$

The belief is thus an information state for control, not merely a regime-classification score.

The actor specifies the portfolio decision conditional on the information state. The critic approximates the conditional expected continuation objective from that state, and actor-critic learning uses the critic signal to improve the actor. Reinforcement learning is employed here as a simulation-based numerical approximation method; stochastic filtering and control theory determine the state representation and restrict the functional forms being learned.

Table 2: Reinforcement-learning objects in the portfolio-control problem.

Object	Stage 1: frictionless control	Stage 2: costly rebalancing
Information state	(t, X_t, p_t)	$(t, X_t, p_t, \mathbf{w}_t)$
Action	Risky-dollar position $\mathbf{u}_t$	Post-trade weight or intervention target
Actor / policy	Gaussian distribution conditional on the belief state	Belief- and horizon-dependent no-trade boundaries
Critic / value model	Expected exploratory mean-variance cost-to-go	QVI value function in the theoretical design; no separate critic in the reported Direct Boundary implementation
Learning objective	Martingale and terminal cost-to-go signals	Differentiable terminal utility plus turnover, gap, and leverage regularization
Learning role	Estimate low-dimensional policy and value coefficients	Estimate boundary geometry around the frozen Stage 1 center

Figure 1 summarizes the information flow. The hidden regime affects observed returns but is not supplied directly to the controller. The filter compresses the observation history into p_t , which enters the state used by the actor or boundary policy.

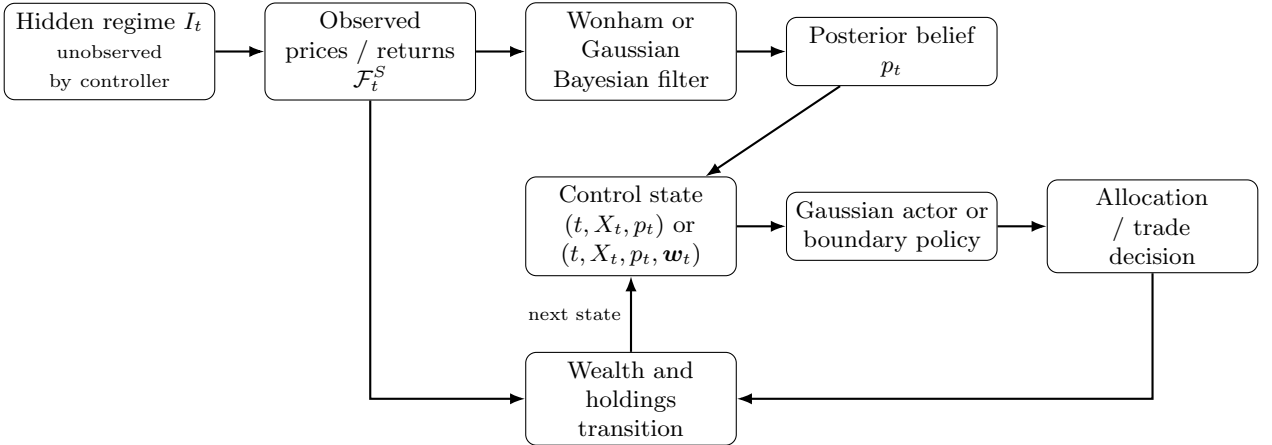


Figure 1: Belief-state construction and portfolio-control information flow.

3.2 Selective use of deep function approximation

Neural networks provide flexible approximations to high-dimensional nonlinear functions, but flexibility is not costless: a larger function class can increase optimization variance, data requirements, computational burden, and the difficulty of understanding the resulting decision rule. The present framework therefore does not use deep learning uniformly across the two stages. Model complexity is allocated according to the extent to which

stochastic-control theory identifies the relevant functional form. Deep learning is not treated as inherently uninterpretable; interpretability depends substantially on what the network is permitted to represent.

In Stage 1, the HJB implies a Gaussian exploratory policy, a policy mean that is linear in wealth relative to the multiplier, and a value function that is quadratic in wealth. A low-dimensional parametrization is therefore used instead of a generic deep state-to-weight network. This choice preserves the analytically identified structure and avoids asking simulation data to relearn a functional form already supplied by the control problem. The principal motivations are identifiability, sample efficiency, and numerical discipline; reduced computation and improved interpretability are additional benefits rather than the primary justification.

In Stage 2, the transaction-cost QVI identifies the enlarged state, continuation-versus-intervention logic, proportional-cost wealth transition, and qualitative role of an inaction region around a frictionless center. It does not characterize the finite-horizon multidimensional lower and upper boundaries sufficiently sharply for the present implementation. The reported neural network is therefore not asked to learn an unrestricted portfolio policy. It learns only the residual time-, belief-, wealth-, and holdings-dependence of the boundaries, subject to center-relative geometry, ordering, and feasibility constraints. The architecture is thus structurally parametric in its economic geometry but flexible in the remaining state dependence.

3.3 Regime-switching market

The latent chain is a parsimonious representation of persistent market conditions, such as favorable versus stressed, risk-on versus risk-off, or normal versus high-volatility environments. It is not a literal claim that financial markets have only two economic states. A continuous-time Markov chain captures random switching while preserving tractability: λ_i is the transition intensity out of regime i , and the expected holding time in that regime is $1/\lambda_i$. The two-state specification is adopted for interpretation and computational economy; additional states, duration dependence, or non-Markov persistence are possible extensions.

The economic labels are determined by the estimated drift and covariance parameters rather than imposed in advance. In the common-covariance benchmark only expected returns change by regime. The later extension also allows volatility and cross-asset correlation to change, reflecting their direct role in diversification and risk concentration.

Let $I_t \in \{1, 2\}$ be a latent continuous-time Markov chain with generator

$$Q = \begin{pmatrix} -\lambda_1 & \lambda_1 \\ \lambda_2 & -\lambda_2 \end{pmatrix}, \quad \lambda_1, \lambda_2 > 0. \quad (2)$$

There are d risky assets and one risk-free asset with rate r . In the common-covariance benchmark,

$$d\mathbf{S}_t = \text{Diag}(\mathbf{S}_t)(\boldsymbol{\mu}_t dt + \sigma d\mathbf{B}_t), \quad \boldsymbol{\mu}_t = \boldsymbol{\mu}^{(1)}\mathbf{1}_{\{I_t=1\}} + \boldsymbol{\mu}^{(2)}\mathbf{1}_{\{I_t=2\}}, \quad (3)$$

where $\sigma \in \mathbb{R}^{d \times d}$ is nonsingular and $\boldsymbol{\Sigma} = \sigma\sigma^\top$. Define

$$\boldsymbol{\rho}^{(i)} = \sigma^{-1}(\boldsymbol{\mu}^{(i)} - r\mathbf{1}), \quad i = 1, 2. \quad (4)$$

If $\mathbf{u}_t \in \mathbb{R}^d$ denotes discounted risky-dollar positions, discounted wealth under full information satisfies

$$dX_t^u = \mathbf{u}_t^\top \sigma(\boldsymbol{\rho}_t dt + d\mathbf{B}_t). \quad (5)$$

The regime is hidden. The investor observes prices and uses the posterior belief

$$p_t = \mathbb{P}(I_t = 1 \mid \mathcal{F}_t^S). \quad (6)$$

The original control problem is therefore a partially observed Markov decision problem. Under the common-covariance assumption, the posterior p_t is a sufficient statistic for the observation history, so the POMDP is converted into a fully observed control problem on the belief state.

Proposition 1 (Belief-state reduction under common covariance). *Suppose the regime follows (2), the asset model is (3), σ is nonsingular and common across regimes, and controls are adapted to the price filtration. Then the innovation process in (8) is Brownian with respect to that filtration, the posterior belief satisfies (11), and discounted wealth satisfies (10). Consequently, (t, X_t, p_t) is a fully observed Markov information state for Stage 1.*

Proof sketch. Define the filtered drift and the innovation process by

$$\hat{\boldsymbol{\mu}}(p) = p\boldsymbol{\mu}^{(1)} + (1-p)\boldsymbol{\mu}^{(2)}, \quad (7)$$

$$d\hat{\mathbf{B}}_t = \sigma^{-1} d \log \mathbf{S}_t - \sigma^{-1} (\hat{\boldsymbol{\mu}}(p_t) - \frac{1}{2} \text{diag}(\boldsymbol{\Sigma})) dt. \quad (8)$$

Because the observation covariance is the same in both regimes, the innovations theorem implies that $\hat{\mathbf{B}}$ is Brownian in the price filtration [17]. Indeed,

$$d \log \mathbf{S}_t = (\boldsymbol{\mu}_t - \frac{1}{2} \text{diag}(\boldsymbol{\Sigma})) dt + \sigma d\mathbf{B}_t = (\hat{\boldsymbol{\mu}}(p_t) - \frac{1}{2} \text{diag}(\boldsymbol{\Sigma})) dt + \sigma d\hat{\mathbf{B}}_t. \quad (9)$$

Substituting this representation into discounted wealth and applying the two-state Wonham filter gives

$$dX_t^u = \mathbf{u}_t^\top \sigma (\hat{\boldsymbol{\rho}}(p_t) dt + d\hat{\mathbf{B}}_t), \quad (10)$$

$$dp_t = a(p_t) dt + b(p_t)^\top d\hat{\mathbf{B}}_t, \quad (11)$$

where

$$\hat{\boldsymbol{\rho}}(p) = p\boldsymbol{\rho}^{(1)} + (1-p)\boldsymbol{\rho}^{(2)}, \quad a(p) = \lambda_2 - (\lambda_1 + \lambda_2)p, \quad b(p) = p(1-p)(\boldsymbol{\rho}^{(1)} - \boldsymbol{\rho}^{(2)}). \quad (12)$$

Thus (X_t, p_t) is the observable state for Stage 1. The shared innovation in (10)–(11) creates a wealth–belief cross-variation term in the HJB. $\square$

For simulation with step Δt , the filter is implemented by a prediction–correction recursion. The Markov prediction is

$$p_{k|k-1} = p_{k-1}(1 - \lambda_1 \Delta t) + (1 - p_{k-1})\lambda_2 \Delta t. \quad (13)$$

Under the common-covariance observation model, the log-return likelihood is

$$y_k \mid I_k = i \sim \mathcal{N}\left(\left(\boldsymbol{\mu}^{(i)} - \frac{1}{2} \text{diag}(\boldsymbol{\Sigma})\right) \Delta t, \boldsymbol{\Sigma} \Delta t\right), \quad (14)$$

and the two predicted log weights are normalized with a log-sum-exp calculation. The controller never observes I_k ; it receives (t_k, X_k, p_k) .

3.4 Exploratory mean–variance objective

For a target expected terminal wealth z , the precommitted mean–variance problem is

$$\min_{\mathbf{u}} \text{Var}(X_T^u) \quad \text{subject to} \quad \mathbb{E}[X_T^u] = z. \quad (15)$$

Here “precommitted” means that the policy is selected at the initial date for the terminal mean–variance criterion and is not subsequently reoptimized as a time-consistent equilibrium strategy. Introducing a Lagrange multiplier ω gives the saddle formulation

$$\sup_{\omega \in \mathbb{R}} \inf_{\mathbf{u}} \left\{ \mathbb{E}[(X_T^u - \omega)^2] - (\omega - z)^2 \right\}. \quad (16)$$

For a fixed multiplier ω , the inner control problem is the entropy-regularized problem developed below. The multiplier is subsequently updated by dual ascent so that the terminal-wealth mean constraint is satisfied; this outer step is given explicitly in (55). Let $\boldsymbol{\pi}(\cdot \mid t, x, p)$ be a stochastic policy density and let $\mathcal{H}(\boldsymbol{\pi}) = -\int \boldsymbol{\pi}(\mathbf{u}) \ln \boldsymbol{\pi}(\mathbf{u}) d\mathbf{u}$ be its differential entropy. The exploratory objective is

$$V^\pi(t, x, p) = \mathbb{E}_{t,x,p} \left[(X_T^\pi - \omega)^2 - \Lambda \int_t^T \mathcal{H}(\boldsymbol{\pi}_s) ds \right] - (\omega - z)^2, \quad (17)$$

and the optimal value is $V^* = \inf_{\boldsymbol{\pi}} V^\pi$. The entropy coefficient $\Lambda > 0$ controls exploration.

4 Stage 1: HJB-Guided Actor–Critic Learning

4.1 Exploratory HJB and Gaussian policy

Assume that σ is nonsingular, admissible policies have finite second moments and finite differential entropy, and the candidate value satisfies the regularity required for Itô’s formula. For a fixed deterministic action $\mathbf{u}$, the observable state dynamics imply

$$d\langle X \rangle_t = \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u} dt, \quad (18)$$

$$d\langle p \rangle_t = \|b(p)\|^2 dt, \quad (19)$$

$$d\langle X, p \rangle_t = \mathbf{u}^\top \sigma b(p) dt. \quad (20)$$

Consequently, the controlled generator applied to a smooth function v is

$$\mathcal{L}^{\mathbf{u}} v = v_t + a(p)v_p + \frac{1}{2}\|b(p)\|^2 v_{pp} + \mathbf{u}^\top \sigma \hat{\rho}(p)v_x \quad (21)$$

$$+ \frac{1}{2}\mathbf{u}^\top \Sigma \mathbf{u} v_{xx} + \mathbf{u}^\top \sigma b(p)v_{xp}. \quad (22)$$

The last term is essential: it is generated by the common innovation driving both wealth and belief. Averaging the action-dependent terms in (22) under a policy density gives the control-dependent part of the exploratory HJB,

$$\min_{\boldsymbol{\pi}} \int_{\mathbb{R}^d} \left[\frac{1}{2}\mathbf{u}^\top \Sigma \mathbf{u} v_{xx} + \mathbf{u}^\top \sigma \hat{\rho}(p)v_x + \mathbf{u}^\top \sigma b(p)v_{xp} + \Lambda \ln \pi(\mathbf{u}) \right] \pi(\mathbf{u}) d\mathbf{u}. \quad (23)$$

The remaining terms are $v_t + a(p)v_p + \frac{1}{2}\|b(p)\|^2 v_{pp}$, and the terminal condition is

$$v(T, x, p) = (x - \omega)^2 - (\omega - z)^2. \quad (24)$$

To solve the minimization over densities, define

$$A = v_{xx} \Sigma, \quad q = \sigma [\hat{\rho}(p)v_x + b(p)v_{xp}]. \quad (25)$$

If $v_{xx} > 0$, then A is positive definite. Let

$$m = -A^{-1}q, \quad \varphi^*(\mathbf{u}) = \mathcal{N}(\mathbf{u} \mid m, \Lambda A^{-1}). \quad (26)$$

Completing the square and using the Gibbs variational identity gives

$$\begin{aligned} & \int_{\mathbb{R}^d} \left[\frac{1}{2}\mathbf{u}^\top A \mathbf{u} + q^\top \mathbf{u} + \Lambda \log \pi(\mathbf{u}) \right] \pi(\mathbf{u}) d\mathbf{u} \\ &= \Lambda \text{KL}(\pi \parallel \varphi^*) - \frac{1}{2}q^\top A^{-1}q - \frac{\Lambda}{2} \log \det(2\pi \Lambda A^{-1}). \end{aligned} \quad (27)$$

The Gibbs identity yields the following pointwise characterization.

Proposition 2 (Gaussian exploratory Hamiltonian optimizer). *If $v_{xx} > 0$ and the policy density has finite second moment and finite differential entropy, the unique minimizer of the entropy-regularized Hamiltonian (23) is*

$$\pi^*(\cdot \mid t, x, p) = \mathcal{N}\left(-\frac{(\sigma^\top)^{-1}[\hat{\rho}(p)v_x + b(p)v_{xp}]}{v_{xx}}, \frac{\Lambda}{v_{xx}} \Sigma^{-1}\right). \quad (28)$$

Proof. Equation (27) writes the objective as a constant plus $\Lambda \text{KL}(\pi \parallel \varphi^*)$. The Kullback–Leibler divergence is nonnegative and vanishes only when $\pi = \varphi^*$ almost everywhere, which gives (28). $\square$

This proposition is the main reduction used by the actor: the policy family is Gaussian by construction, and learning focuses on its low-dimensional mean and covariance coefficients.

Completing the square also gives the optimized HJB explicitly:

$$\begin{aligned} 0 &= v_t + a(p)v_p + \frac{1}{2}\|b(p)\|^2 v_{pp} - \frac{\|\hat{\rho}(p)v_x + b(p)v_{xp}\|^2}{2v_{xx}} \\ &\quad - \frac{\Lambda}{2} \log \det\left(\frac{2\pi\Lambda}{v_{xx}} \Sigma^{-1}\right). \end{aligned} \quad (29)$$

The first negative term is the minimized quadratic Hamiltonian and the logarithmic term is the entropy contribution evaluated at the Gaussian density.

4.2 Quadratic value representation

The terminal condition motivates

$$v(t, x, p) = \frac{(x - \omega)^2}{f(t, p)} + g(t, p) - (\omega - z)^2, \quad f(t, p) > 0. \quad (30)$$

Then

$$v_x = \frac{2(x - \omega)}{f}, \quad v_{xx} = \frac{2}{f}, \quad v_{xp} = -2(x - \omega) \frac{f_p}{f^2}. \quad (31)$$

The remaining derivatives needed for coefficient matching are

$$v_t = -(x - \omega)^2 \frac{f_t}{f^2} + g_t, \quad (32)$$

$$v_p = -(x - \omega)^2 \frac{f_p}{f^2} + g_p, \quad (33)$$

$$v_{pp} = -(x - \omega)^2 \frac{f_{pp}}{f^2} + 2(x - \omega)^2 \frac{f_p^2}{f^3} + g_{pp}. \quad (34)$$

Substitution into (28) yields

$$\pi^*(\cdot \mid t, x, p) = \mathcal{N} \left(-(x - \omega)(\sigma^\top)^{-1} \left[\hat{\rho}(p) - b(p) \frac{f_p(t, p)}{f(t, p)} \right], \frac{\Lambda f(t, p)}{2} \Sigma^{-1} \right). \quad (35)$$

The policy mean separates into a myopic demand and an intertemporal hedging demand:

$$\begin{aligned} \mathbb{E}[\mathbf{u}_t \mid t, x, p] = & -(x - \omega)(\sigma^\top)^{-1} \hat{\rho}(p) \\ & + (x - \omega)(\sigma^\top)^{-1} b(p) \frac{f_p(t, p)}{f(t, p)}. \end{aligned} \quad (36)$$

The first component responds to the current filtered Sharpe ratio, whereas the second hedges changes in future investment opportunities induced by belief fluctuations. The exploration covariance is proportional to $\Lambda f(t, p)$ and inversely proportional to the return covariance. As $\Lambda \downarrow 0$, the Gaussian policy converges weakly to a point mass at its mean, recovering the classical feedback control under the same regularity conditions.

The Gaussian family is also supported by policy improvement. Given the value V^π of an admissible policy, replacing that policy pointwise by the Gaussian obtained from (28) with $v = V^\pi$ minimizes the local exploratory Hamiltonian. If the updated policy remains admissible, the standard verification argument gives $V^{\pi'} \leq V^\pi$ under the cost-minimization convention. Thus restricting learning to structured Gaussian actors is not merely a convenient distributional assumption.

The first two moments of this distribution are respectively linear and quadratic in $(x - \omega)$, so the HJB remains closed within the wealth-quadratic class. The original PDE in (t, x, p) therefore reduces to coefficient equations in (t, p) .

Proposition 3 (Wealth-quadratic closure). *Under the assumptions of Proposition 2, suppose a sufficiently regular solution of the optimized HJB has the form (30) with $f > 0$. Then the Gaussian actor is given by (35), and the HJB closes in the two coefficient functions f and g , which solve (38)–(39). In particular, the equation for f is linear and independent of g .*

Proof sketch. To see the reduction explicitly, substitute (30) into (29) and collect the coefficient of $(x - \omega)^2$. The terms involving f_p^2/f cancel because the exact wealth–belief cross coefficient satisfies

$$(\sigma b)^\top \Sigma^{-1} (\sigma b) = \|b\|^2. \quad (37)$$

The function f therefore solves the linear parabolic PDE

$$f_t + [a(p) - 2\hat{\rho}(p)^\top b(p)] f_p + \frac{1}{2} \|b(p)\|^2 f_{pp} + \|\hat{\rho}(p)\|^2 f = 0, \quad f(T, p) = 1. \quad (38)$$

The constant coefficient gives

$$g_t + a(p)g_p + \frac{1}{2} \|b(p)\|^2 g_{pp} - \frac{\Lambda}{2} \log \det (\pi \Lambda f(t, p) \Sigma^{-1}) = 0, \quad g(T, p) = 0. \quad (39)$$

Equations (38)–(39) are the exact lower-dimensional coefficient system for the common-covariance benchmark, subject to the regularity and admissibility conditions required by the verification theorem. The equation for f is independent of g , while g is solved after f . $\square$

Table 3 distinguishes structures identified by filtering or control theory from quantities that remain to be estimated numerically. This separation is the operational meaning of *structure-guided* learning in the paper.

Table 3: Control-theoretic structure and learned quantities.

Component	Structure supplied by theory	Quantity estimated numerically
Information state	Posterior belief is sufficient under the filtering model	Regime parameters and recursive belief path
Stage 1 actor	Gaussian family; mean linear in $x - \omega$; covariance proportional to $f \Sigma^{-1}$	Low-dimensional belief/time coefficients and relaxed covariance coefficients
Stage 1 critic	Quadratic dependence on wealth	Coefficient surfaces $f(t, p)$ and $g(t, p)$
Stage 2 policy	QVI continuation/intervention logic and existence of an inaction region	Lower and upper boundary geometry around the Stage 1 center
Transaction costs	Self-financing wealth recursion with proportional cost	Neural boundary parameters under the pathwise objective

The critic uses polynomial coefficient surfaces,

$$\ell_\theta(t, p) = \log f_\theta(t, p) = \sum_{i=0}^m \sum_{j=0}^m \theta_{ij} p^i (T - t)^j, \quad (40)$$

$$g_\theta(t, p) = \sum_{i=0}^{m_g} \sum_{j=0}^{m_g} \tilde{\theta}_{ij} p^i (T - t)^j, \quad (41)$$

and

$$V_\theta(t, x, p; \omega) = \frac{(x - \omega)^2}{\exp(\ell_\theta(t, p))} + g_\theta(t, p) - (\omega - z)^2. \quad (42)$$

The exponential parametrization preserves $f_\theta > 0$. The actor mirrors (35), with a small number of trainable coefficients that allow modest deviations from the exact formula to absorb approximation and filtering errors.

The derivatives used in critic optimization are available analytically. If $\psi_{ij}(t, p) = p^i (T - t)^j$, then

$$\frac{\partial V_\theta}{\partial \theta_{ij}} = -\frac{(x - \omega)^2}{f_\theta(t, p)} \psi_{ij}(t, p), \quad \frac{\partial V_\theta}{\partial \tilde{\theta}_{ij}} = \psi_{ij}(t, p). \quad (43)$$

The displayed polynomial basis does not automatically impose the terminal conditions. If hard terminal consistency is desired, one can instead parameterize

$$\ell_\theta(t, p) = (T - t) \tilde{\ell}_\theta(t, p), \quad g_\theta(t, p) = (T - t) \tilde{g}_\theta(t, p), \quad (44)$$

which guarantees $f_\theta(T, p) = 1$ and $g_\theta(T, p) = 0$; alternatively, the terminal equations can be imposed through supervised residual penalties.

More concretely, the code fixes the wealth dependence through $(x - \omega)$ and learns only low-dimensional belief/time coefficients. A representative actor is

$$\pi_\phi(\mathbf{u} \mid t, x, p) = \mathcal{N}\left(-(x - \omega) \alpha_\phi(p, \partial_p \ell_\theta(t, p)), \frac{\Lambda}{2} f_\theta(t, p) \Sigma_\phi(p)\right), \quad (45)$$

where $\Sigma_\phi(p)$ is constrained to be positive definite. In the exact benchmark, α_ϕ and Σ_ϕ reduce to the coefficients in (35); the relaxed implementation allows small departures without discarding the HJB-implied Gaussian and wealth-linear structure.

A concrete low-dimensional mean map consistent with the exact formula is

$$\alpha_\phi(p, \partial_p \ell_\theta) = (\sigma^\top)^{-1} [\phi_1 p + \phi_2 - \phi_3 p(1 - p) \partial_p \ell_\theta(t, p)], \quad (46)$$

where the vector coefficients ϕ_1, ϕ_2, ϕ_3 learn the filtered myopic and hedging directions. A Cholesky or matrix-exponential parametrization of Σ_ϕ preserves positive definiteness throughout training. The resulting number of actor parameters grows with the asset dimension but not with an unrestricted discretization of the wealth state.

4.3 Policy evaluation and improvement

At each discrete time $t_k = k\Delta t$, the state is $s_k = (t_k, X_k, p_k)$, an action is sampled from the current Gaussian actor, and frictionless wealth is propagated by

$$\mathbf{u}_k \sim \pi_\phi(\cdot \mid s_k), \quad X_{k+1} = X_k + \mathbf{u}_k^\top (\exp(\Delta \log \mathbf{S}_k - r\Delta t) - \mathbf{1}), \quad (47)$$

where the exponential is componentwise. The observed return is then passed to the appropriate common- or regime-dependent-covariance filter.

For a fixed policy, V^π satisfies a linear policy-evaluation PDE. Applying Itô's formula and using that PDE shows that

$$M_t = V^\pi(t, X_t, p_t) - \Lambda \int_0^t \mathcal{H}_s ds \quad (48)$$

is a martingale. Hence

$$V^\pi(t_k, X_k, p_k) = \mathbb{E}_k \left[(X_T - \omega)^2 - (\omega - z)^2 - \Lambda \int_{t_k}^T \mathcal{H}_s ds \right]. \quad (49)$$

This is the continuous-time justification for the terminal Monte Carlo target below; it is not an arbitrary supervised label.

For a simulated episode $\{(t_k, X_k, p_k, \mathbf{u}_k)\}_{k=0}^K$, define the realized cost-to-go

$$G_k = (X_T - \omega)^2 - (\omega - z)^2 - \Lambda \sum_{i=k}^{K-1} \mathcal{H}_i \Delta t. \quad (50)$$

The critic minimizes the Monte Carlo value error

$$\mathcal{L}_{\text{critic}}(\theta) = \frac{1}{2} \mathbb{E}[(V_\theta(t_k, X_k, p_k) - G_k)^2]. \quad (51)$$

The local martingale increment is

$$\Delta M_k = V_\theta(t_{k+1}, X_{k+1}, p_{k+1}) - V_\theta(t_k, X_k, p_k) - \Lambda \mathcal{H}_k \Delta t. \quad (52)$$

In implementation, the actor signal mixes a normalized local increment with the full-horizon tail cost residual

$$r_k^{\text{tail}} = G_k - V_\theta(t_k, X_k, p_k), \quad s_k = (1 - \eta) \text{Norm}(\Delta M_k) + \eta \text{Norm}(r_k^{\text{tail}}). \quad (53)$$

Because s_k is a cost signal, its sign in the policy loss is chosen consistently with minimization. The tail term is an implementation refinement for finite-sample credit assignment; the martingale relation remains the theoretical anchor.

Writing $\text{sg}(\cdot)$ for a stop-gradient operation, a cost-minimizing score-function surrogate is

$$\mathcal{L}_{\text{actor}}(\phi) = \mathbb{E}[\log \pi_\phi(\mathbf{u}_k | t_k, X_k, p_k) \text{sg}(s_k) - \Lambda \mathcal{H}(\pi_\phi(\cdot | t_k, X_k, p_k)) \Delta t]. \quad (54)$$

Positive values of the cost signal reduce the probability of the sampled action under gradient descent. The separate entropy derivative in (54) is required because the entropy depends directly on ϕ and is not recovered by stopping the gradient through s_k . Under a reward convention, both the signal and the optimization sign must be reversed consistently.

The implementation permits one or more critic updates per outer iteration and normalizes the martingale and tail signals within a batch. The reported configurations use one critic step. The pure martingale-difference update is closest to continuous-time theory but was empirically slow and noisy; the mixed signal with weight $\eta = 0.3$ provided a stronger full-horizon correction in the representative runs. This modification is algorithmic rather than a new control theorem.

The multiplier is updated through the sample terminal-wealth constraint,

$$\omega \leftarrow \omega - \alpha_\omega (\bar{X}_T - z). \quad (55)$$

Here $\bar{X}_T$ is the batch estimate of expected terminal wealth, optionally smoothed by an exponential moving average; the reported configurations use smoothing parameter 0.5. Indeed, differentiating the Lagrangian value with respect to ω gives a quantity proportional to $-(\mathbb{E}[X_T] - z)$; the outer step is therefore a dual-ascent update for enforcing the target mean.

Algorithm 1 summarizes the complete Stage 1 loop. Regime parameters are estimated before training, but beliefs are updated recursively inside every episode from observed returns only.

Algorithm 1 Stage 1 HJB-guided actor–critic under partial information

```
1: Estimate  $(\boldsymbol{\mu}^{(i)}, \boldsymbol{\Sigma}^{(i)}, \lambda_i)$  and initialize  $\theta$ ,  $\phi$ , and  $\omega$ .
2: for outer iteration  $m = 1, \dots, M$  do
3:   Simulate or sample return paths and initialize  $(X_0, p_0)$ .
4:   for  $k = 0, \dots, K - 1$  do
5:     Sample  $\mathbf{u}_k \sim \boldsymbol{\pi}_\phi(\cdot \mid t_k, X_k, p_k)$ .
6:     Propagate  $X_{k+1}$  using (47).
7:     Update  $p_{k+1}$  by the prediction–correction filter using the observed return.
8:   end for
9:   Construct all tail targets  $G_k$  from (50).
10:  Perform one or more critic steps on  $\mathcal{L}_{\text{critic}}$ .
11:  Form  $\Delta M_k$ ,  $r_k^{\text{tail}}$ , and the normalized mixed signal  $s_k$ .
12:  Update the Gaussian actor using (54).
13:  Update  $\omega$  using (55).
14: end for
```

5 Regime Estimation and Filtering in Practice

5.1 Legacy threshold labeling

The filter recursion is theory-driven, but its inputs must be estimated. The original heuristic route constructs a bull/bear path from an asset or equal-weight index. If P_t is the index level, the rule switches from bull to bear when

$$P_t \leq P_t^{\text{peak}}(1 - Y_l), \quad (56)$$

and from bear to bull when

$$P_t \geq P_t^{\text{trough}}(1 + Y_u). \quad (57)$$

In the legacy configuration, $Y_l = 0.19$ and $Y_u = 0.24$ are fixed hyperparameters rather than cross-validated estimates. This method is transparent but sensitive to threshold selection and can produce unreliable labels in a multi-asset setting.

5.2 Statistical jump model

The current regime-dependent-covariance experiment instead uses an equal-weight return index and constructs a standardized feature vector z_k from rolling means and volatilities, downside deviation, rolling drawdown, short-versus-long mean spreads, and volatility ratios. Every rolling feature is intended to use information available at or before time k only. A two-state statistical jump model jointly estimates regime representatives ϑ_1, ϑ_2 and a persistent label path $s_{1:n}$ by solving a problem of the form

$$\min_{\vartheta_1, \vartheta_2, s_{1:n}} \sum_{k=1}^n \|z_k - \vartheta_{s_k}\|_2^2 + \gamma_J \sum_{k=2}^n \mathbf{1}_{\{s_k \neq s_{k-1}\}}. \quad (58)$$

The first term clusters economically similar feature observations; the jump penalty γ_J suppresses implausibly rapid switching. The reported configuration uses a separate 20-year estimation sample, $\gamma_J = 100$, 10 random initializations, and at most 400 iterations. The estimation sample is separated from the independently generated policy-evaluation paths.

The archived configuration records the estimation horizon and optimization settings, but it does not retain the exact rolling windows, feature-by-feature normalization statistics, or the random seed used to generate the estimation sample. Those items must be recorded in the frozen final manifest. In particular, the final implementation record will specify each lookback window, confirm that standardization statistics are computed from the estimation sample only, and document the separation among SJM fitting, policy training, and held-out simulation evaluation.

The discrete SJM is fitted by block coordinate descent, as reviewed by Aydinhan et al. [1]. For a fixed state path, the regime representatives solve

$$\vartheta_j^{(m+1)} = \arg \min_{\vartheta} \sum_{k: s_k^{(m)} = j} \|z_k - \vartheta\|_2^2, \quad (59)$$

so under squared Euclidean loss each representative is the sample mean of the feature vectors assigned to that regime. For fixed representatives, the entire label path is optimized jointly rather than greedily. Define the Bellman cost

$$C_k(j) = \min_{s_{1:k-1}} \left\{ \sum_{r=1}^k \|z_r - \vartheta_{s_r}\|_2^2 + \gamma_J \sum_{r=2}^k \mathbf{1}_{\{s_r \neq s_{r-1}\}} : s_k = j \right\}. \quad (60)$$

It satisfies the dynamic-programming recursion

$$C_1(j) = \|z_1 - \vartheta_j\|_2^2, \quad (61)$$

$$C_k(j) = \|z_k - \vartheta_j\|_2^2 + \min_{i \in \{1, \dots, K\}} \{C_{k-1}(i) + \gamma_J \mathbf{1}_{\{i \neq j\}}\}. \quad (62)$$

Storing the minimizing predecessor at every (k, j) and backtracking from $\arg \min_j C_n(j)$ recovers the globally optimal state sequence conditional on the current representatives. The recursion costs $O(nK^2)$ time and $O(nK)$ memory with full backpointers; for the two-state model used here, its time complexity is linear in the sample length.

Each coordinate-descent step weakly decreases the SJM objective, but the joint problem is nonconvex because labels and representatives are optimized together. The algorithm therefore guarantees only a coordinate-wise stationary solution, not a global optimum. Multiple K-means++-type initializations are used and the solution with the lowest objective is retained. Thus, dynamic programming solves the discrete path subproblem exactly, while random restarts address—but do not eliminate—the global nonconvexity of the complete fitting problem.

Given inferred labels $\hat{I}_k$, the parameters are estimated as follows:

- $\hat{\mu}^{(i)}$ is the annualized mean return within regime i ;
- $\hat{\Sigma}^{(i)}$ is the annualized log-return covariance within regime i ;
- $\hat{\lambda}_1$ and $\hat{\lambda}_2$ are obtained from empirical transition frequencies or, equivalently, inverse average sojourn times.

The jump model is not the online regime predictor used by the controller. It provides a persistent and practically usable segmentation for parameter estimation. Online beliefs are subsequently updated by the Wonham or discrete-time Gaussian Bayes filter, preserving the belief-state HJB architecture.

If N_{ij} is the number of inferred transitions from state i to state j , the empirical transition matrix is

$$\hat{P}_{ij} = \frac{N_{ij}}{\sum_{j'} N_{ij'}}, \quad (63)$$

and the continuous-time switching intensities are obtained from the small-step approximation $\hat{\lambda}_1 \simeq \hat{P}_{12}/\Delta t$ and $\hat{\lambda}_2 \simeq \hat{P}_{21}/\Delta t$, or equivalently from inverse mean sojourn times.

Table 4 reports the SJM-derived parameters used in the regime-dependent run. Regime 1 has positive mean returns and lower covariance, whereas regime 2 has negative mean returns, higher marginal volatility, and higher cross-asset correlation. The estimated switching process is less asymmetric than the true simulation process: it exits regime 1 more frequently and exits regime 2 less rapidly.

Table 4: Statistical jump model results used by the regime-dependent-covariance filter.

Quantity	Regime 1	Regime 2
Annualized mean return	(0.3830, 0.2759)	(−0.3243, −0.1476)
Estimated covariance	$\begin{pmatrix} 0.0485 & 0.0108 \\ 0.0108 & 0.0322 \end{pmatrix}$	$\begin{pmatrix} 0.0574 & 0.0193 \\ 0.0193 & 0.0401 \end{pmatrix}$
Annualized volatilities	(0.220, 0.179)	(0.240, 0.200)
Cross-asset correlation	0.274	0.402
Leaving intensity	$\hat{\lambda}_1 = 0.6801$	$\hat{\lambda}_2 = 1.3310$
Mean sojourn time	1.47 years (370 days)	0.75 years (189 days)

The stationary mass implied by the estimated rates is

$$\hat{p}_1 = \frac{\hat{\lambda}_2}{\hat{\lambda}_1 + \hat{\lambda}_2} \approx 0.662. \quad (64)$$

This estimate is materially below the true simulation value 0.889, which helps explain the noisier belief and learning dynamics in the regime-dependent experiment.

At discrete times, prediction under (2) is

$$p_{k|k-1} = p_{k-1}(1 - \lambda_1 \Delta t) + (1 - p_{k-1})\lambda_2 \Delta t. \quad (65)$$

The predicted weights are combined with regime-wise Gaussian likelihoods and normalized in log space. This distinction between *parameter estimation* and *online filtering* permits richer offline regime identification without replacing the posterior state required by the control model.

6 Regime-Dependent Covariance Extension

6.1 Discrete-time Gaussian filtering layer

In the extension, both drift and covariance depend on the regime:

$$\Delta \log \mathbf{S}_k \mid I_k = i \sim \mathcal{N} \left(\left(\boldsymbol{\mu}^{(i)} - \frac{1}{2} \text{diag}(\boldsymbol{\Sigma}^{(i)}) \right) \Delta t, \boldsymbol{\Sigma}^{(i)} \Delta t \right). \quad (66)$$

Let $\phi_i(y_k)$ denote the Gaussian density in (66). After the Markov prediction (13), Bayes' rule gives

$$p_k = \frac{p_{k|k-1} \phi_1(y_k)}{p_{k|k-1} \phi_1(y_k) + (1 - p_{k|k-1}) \phi_2(y_k)}. \quad (67)$$

In computation, the numerator and denominator are evaluated through normalized log weights. The Bayes correction in (67) is exact conditional on the specified Gaussian emission model and the predicted probability. The prediction (13), however, is the first-order Euler discretization of the continuous-time Markov chain; an exactly discretized alternative would use the transition matrix $\exp(Q\Delta t)$.

The continuous-time separation structure is more delicate because the hidden state affects the observation quadratic variation as well as its drift. In particular, the common- σ innovation construction (8) cannot be reused without a new filtering theorem. The extension therefore separates a discrete-time prediction–correction filtering layer from an approximate control layer.

6.2 Moment-matched effective diffusion

Write $m_i = \boldsymbol{\mu}^{(i)} - \frac{1}{2} \text{diag}(\boldsymbol{\Sigma}^{(i)})$. Conditional on a prior belief p , the one-step mixture has

$$\mathbb{E}[\Delta \log \mathbf{S}_k \mid p] = (pm_1 + (1 - p)m_2) \Delta t, \quad (68)$$

and

$$\begin{aligned} \text{Cov}(\Delta \log \mathbf{S}_k \mid p) &= \left[p\boldsymbol{\Sigma}^{(1)} + (1 - p)\boldsymbol{\Sigma}^{(2)} \right] \Delta t \\ &\quad + p(1 - p)(m_1 - m_2)(m_1 - m_2)^\top (\Delta t)^2. \end{aligned} \quad (69)$$

Thus the belief-averaged covariance is the leading-order $O(\Delta t)$ conditional covariance; the between-regime drift-mixture term is $O((\Delta t)^2)$. For control, the implementation defines

$$\hat{\mathbf{b}}(p) = p(\boldsymbol{\mu}^{(1)} - r\mathbf{1}) + (1 - p)(\boldsymbol{\mu}^{(2)} - r\mathbf{1}), \quad \bar{\boldsymbol{\Sigma}}(p) = p\boldsymbol{\Sigma}^{(1)} + (1 - p)\boldsymbol{\Sigma}^{(2)}. \quad (70)$$

The controlled wealth process is approximated by the moment-matched diffusion

$$dX_t \approx \mathbf{u}_t^\top \hat{\mathbf{b}}(p_t) dt + \mathbf{u}_t^\top \bar{\boldsymbol{\sigma}}(p_t) d\mathbf{B}_t, \quad \bar{\boldsymbol{\sigma}}(p) \bar{\boldsymbol{\sigma}}(p)^\top = \bar{\boldsymbol{\Sigma}}(p). \quad (71)$$

Equation (71) is a single effective diffusion that matches the leading conditional drift and covariance. It is not the exact separated wealth dynamics of the original partially observed continuous-time model.

6.3 Approximate HJB and Gaussian actor

To make the approximation explicit, posit effective belief dynamics

$$dp_t = \alpha(p_t) dt + \beta(p_t)^\top dW_t, \quad q(p) = \|\beta(p)\|^2, \quad (72)$$

and let $\tilde{c}(p) \in \mathbb{R}^d$ denote the effective covariance between wealth noise per unit action and belief noise. The formal approximate HJB is

$$0 = v_t + \alpha(p)v_p + \frac{1}{2}q(p)v_{pp} + \min_{\pi} \int_{\mathbb{R}^d} \left[\frac{1}{2} \mathbf{u}^\top \bar{\Sigma}(p) \mathbf{u} v_{xx} + \mathbf{u}^\top (\hat{\mathbf{b}}(p)v_x + \tilde{c}(p)v_{xp}) + \Lambda \ln \pi(\mathbf{u}) \right] \pi(\mathbf{u}) d\mathbf{u}. \quad (73)$$

The Hamiltonian is still quadratic in $\mathbf{u}$. Therefore, if $v_{xx} > 0$,

$$\pi^*(\cdot | t, x, p) = \mathcal{N} \left(-\bar{\Sigma}(p)^{-1} \frac{\hat{\mathbf{b}}(p)v_x + \tilde{c}(p)v_{xp}}{v_{xx}}, \frac{\Lambda}{v_{xx}} \bar{\Sigma}(p)^{-1} \right). \quad (74)$$

With the same wealth-quadratic ansatz (30), let $\ell = f_p/f$. The policy becomes

$$\pi^* = \mathcal{N} \left(-(x - \omega) \bar{\Sigma}(p)^{-1} [\hat{\mathbf{b}}(p) - \tilde{c}(p)\ell(t, p)], \frac{\Lambda f(t, p)}{2} \bar{\Sigma}(p)^{-1} \right). \quad (75)$$

Collecting the $(x - \omega)^2$ terms in (73) yields the approximate coefficient equation

$$0 = f_t + \left[\alpha(p) - 2\hat{\mathbf{b}}(p)^\top \bar{\Sigma}(p)^{-1} \tilde{c}(p) \right] f_p + \frac{1}{2}q(p)f_{pp} + f\hat{\mathbf{b}}(p)^\top \bar{\Sigma}(p)^{-1} \hat{\mathbf{b}}(p) + [\tilde{c}(p)^\top \bar{\Sigma}(p)^{-1} \tilde{c}(p) - q(p)] \frac{f_p^2}{f}, \quad f(T, p) = 1. \quad (76)$$

The constant term gives

$$g_t + \alpha(p)g_p + \frac{1}{2}q(p)g_{pp} - \frac{\Lambda}{2} \log \det (\pi \Lambda f(t, p) \bar{\Sigma}(p)^{-1}) = 0, \quad g(T, p) = 0. \quad (77)$$

Equation (76) reveals the precise difference from the theorem-level benchmark. In the common-covariance case,

$$\tilde{c} = \sigma b, \quad \bar{\Sigma} = \sigma \sigma^\top, \quad q = \|b\|^2, \quad (78)$$

so $\tilde{c}^\top \bar{\Sigma}^{-1} \tilde{c} = q$ and the nonlinear f_p^2/f term cancels, recovering (38). Under the regime-dependent approximation, this identity is not established; the additional nonlinear term must be retained unless the effective coefficients are deliberately constrained to satisfy it.

In the reported implementation, α , β , and the wealth–belief cross coefficient $\tilde{c}$ are not separately identified from a continuous-time regime-dependent-covariance filtering model. Beliefs are updated by the Euler-prediction and exact Gaussian Bayes-correction recursion in (67), and the belief-averaged covariance $\bar{\Sigma}(p)$ is imposed explicitly in the actor. The remaining effects represented formally by α , β , and $\tilde{c}$ are absorbed into the relaxed low-dimensional actor and critic coefficients. Consequently, (73)–(76) motivate the Gaussian architecture and coefficient dependence; they are not PDEs solved directly in the reported experiment.

6.4 Exact and approximate components

- **Exact conditional correction:** the Gaussian Bayes update in (67), conditional on the predicted probability and specified emission densities.
- **Time-discretization convention:** the reported Markov prediction uses the first-order formula (13), rather than the exact matrix exponential $\exp(Q\Delta t)$.
- **Moment-matched approximation:** replacement of the posterior Gaussian mixture by (71).
- **Structurally preserved:** quadratic Hamiltonian, Gaussian exploratory policy, wealth-quadratic ansatz, and low-dimensional coefficient representation.
- **Implementation convention:** $\bar{\Sigma}(p)$ and the discrete Bayesian belief recursion are explicit, while the continuous-time coefficients $(\alpha, \beta, \tilde{c})$ are not estimated separately.
- **Not yet proved:** an exact continuous-time filter with regime-dependent covariance, a separation principle, identification of $\tilde{c}$, admissibility and well-posedness of (76)–(77), and a verification theorem.

Remark 1. *The regime-dependent results should therefore be interpreted as a discretized prediction–correction filter combined with a theory-guided approximate controller. The numerical experiment tests whether this architecture remains learnable; it does not establish optimality for the original continuous-time partially observed model.*

7 Stage 2: Transaction-Cost-Aware No-Trade Learning

7.1 Transaction-cost QVI benchmark

With transaction costs, the current position affects all future trading costs. The observable state must therefore be enlarged from (t, x, p) to $(t, x, p, \mathbf{w})$. In the no-trade region, consider the weights-based diffusion

$$dX_t = X_t \bar{\mu}(p_t, \mathbf{w}_t) dt + X_t \bar{\sigma}(p_t, \mathbf{w}_t)^\top d\hat{\mathbf{B}}_t, \quad (79)$$

$$dp_t = \mu_p(p_t) dt + \beta(p_t)^\top d\hat{\mathbf{B}}_t, \quad (80)$$

$$d\mathbf{w}_t = b_w(p_t, \mathbf{w}_t) dt + D(p_t, \mathbf{w}_t) d\hat{\mathbf{B}}_t. \quad (81)$$

The continuation generator $\mathcal{L}$ contains all state covariations generated by the common innovation. In particular,

$$a_{Xp} = x\bar{\sigma}^\top \beta, \quad a_{Xw} = xD\bar{\sigma}, \quad a_{pw} = D\beta, \quad a_{ww} = DD^\top. \quad (82)$$

Let $\mathcal{M}$ be the intervention operator. For a cost-minimization convention, the transaction-cost dynamic program is the QVI

$$\min\{-V_t - \mathcal{L}V, \mathcal{M}V - V\} = 0. \quad (83)$$

The continuation region satisfies $-V_t - \mathcal{L}V = 0$ and $V < \mathcal{M}V$; intervention occurs on the contact set $V = \mathcal{M}V$. For an approximate critic, a negative obstacle residual $\mathcal{M}_\phi V_\theta - V_\theta$ indicates that the candidate intervention has lower cost than continuation.

Motivated by the Stage 1 terminal condition, consider

$$V(t, x, p, \mathbf{w}) = A(t, p, \mathbf{w})(x - \omega)^2 + B(t, p, \mathbf{w})(x - \omega) + C(t, p, \mathbf{w}), \quad (84)$$

with

$$A(T, p, \mathbf{w}) = 1, \quad B(T, p, \mathbf{w}) = 0, \quad C(T, p, \mathbf{w}) = -(\omega - z)^2. \quad (85)$$

Writing $y = x - \omega$, every derivative of (84) is polynomial in y . Consequently,

$$-V_t - \mathcal{L}V = \Phi_2(t, p, \mathbf{w})y^2 + \Phi_1(t, p, \mathbf{w})y + \Phi_0(t, p, \mathbf{w}), \quad (86)$$

and continuation requires $\Phi_2 = \Phi_1 = \Phi_0 = 0$. Unlike frictionless Stage 1, the linear coefficient B is generally generated by the wealth-proportional dynamics and cannot be fixed at zero away from maturity. Explicit forms of the three coefficients are collected in Appendix A.

7.2 Intervention operator and quadratic closure

Let a general intervention actor choose

$$\mathbf{w}' = \Gamma_\phi(t, p, \mathbf{w}), \quad (87)$$

and let $\kappa(\mathbf{w}, \mathbf{w}')$ be the proportional fraction of wealth lost in trading. Then

$$x^+ = x(1 - \kappa), \quad (\mathcal{M}_\phi V)(t, x, p, \mathbf{w}) = V(t, x^+, p, \Gamma_\phi(t, p, \mathbf{w})). \quad (88)$$

Since $x^+ - \omega = (1 - \kappa)y - \kappa\omega$, the intervention operator preserves the quadratic family whenever Γ_ϕ and κ are independent of x conditional on weights:

$$\mathcal{M}_\phi V = M_2^\phi y^2 + M_1^\phi y + M_0^\phi, \quad (89)$$

where, with $A^\Gamma = A(t, p, \Gamma_\phi)$ and similarly for B^Γ, C^Γ ,

$$M_2^\phi = A^\Gamma(1 - \kappa)^2, \quad (90)$$

$$M_1^\phi = -2A^\Gamma \kappa\omega(1 - \kappa) + B^\Gamma(1 - \kappa), \quad (91)$$

$$M_0^\phi = A^\Gamma \kappa^2 \omega^2 - B^\Gamma \kappa\omega + C^\Gamma. \quad (92)$$

This closure suggests a QVI-guided critic for (A, B, C) and an intervention actor Γ_ϕ .

To retain Stage 1-style low dimensionality in the exploratory QVI design, introduce an exogenous normalized reference map $\mathbf{w}^{\text{ref}}(t, p)$, for example the Stage 1 policy evaluated at a reference wealth level, and define

$$\tau = T - t, \quad \delta = \mathbf{w} - \mathbf{w}^{\text{ref}}(t, p). \quad (93)$$

A structured critic can use even powers of $\|\delta\|_{M(p)}$ for A and C , and odd terms for B ; a corresponding intervention actor can contract δ toward zero. Appendix B gives the proposed basis and residual losses.

This reference-map convention is an architectural approximation. The implemented Stage 1 center weight is generally wealth dependent because $\mathbf{w}^*(t, x, p) = \mathbf{u}^*(t, x, p)/x$. No exact QVI closure is claimed after replacing that map by the wealth-independent reference $\mathbf{w}^{\text{ref}}(t, p)$.

7.3 Exploratory QVI-guided Stage 2 RL design

This subsection records the exploratory actor-critic design considered before adopting Direct Boundary RL. It is a work-in-progress numerical formulation, not a completed verification result. The purpose is to preserve in Stage 2 the same principle used successfully in Stage 1: derive a restricted critic and actor from the control equation before introducing learning.

The critic learns the three coefficient surfaces in (84),

$$A_\theta(\tau, p, \delta), \quad B_\theta(\tau, p, \delta), \quad C_\theta(\tau, p, \delta), \quad (94)$$

where $\tau = T - t$ and $\delta = \mathbf{w} - \mathbf{w}^{\text{ref}}(t, p)$. Positivity of A_θ can be imposed exponentially, while low-order even and odd center-relative bases reflect the quadratic continuation equations. The terminal condition may be enforced either exactly through the parametrization or softly through a terminal residual penalty.

The intervention actor proposes a post-trade target through

$$\Gamma_\phi(t, p, \mathbf{w}) = \mathbf{w} + G_\phi(\tau, p, \delta) = \mathbf{w}^{\text{ref}}(t, p) + \delta + G_\phi(\tau, p, \delta). \quad (95)$$

A low-dimensional radial-directional decomposition is

$$G_\phi(\tau, p, \delta) = -s_\phi(\tau, p, \delta)\delta + R_\phi(\tau, p, \delta), \quad 0 \leq s_\phi \leq 1. \quad (96)$$

With $R_\phi \equiv 0$, $s_\phi = 0$ leaves the position unchanged and $s_\phi = 1$ moves it to the normalized reference center. A simple structured shrinkage factor is

$$s_\phi = \text{sigmoid} \left(\eta_0(\tau, p) + \eta_2(\tau, p) \|\delta\|_{M(p)}^2 + \eta_4(\tau, p) \|\delta\|_{M(p)}^4 \right). \quad (97)$$

The directional correction R_ϕ can be introduced only if the radial model is too restrictive. A final differentiable projection or normalization layer enforces $\Gamma_\phi(t, p, \mathbf{w}) \in \mathcal{W}$ under box, simplex, or leverage constraints.

For the cost-minimization convention, define

$$R_{\text{cont}} = -V_{\theta,t} - \mathcal{L}V_\theta, \quad R_{\text{obs}} = \mathcal{M}_\phi V_\theta - V_\theta. \quad (98)$$

The exact QVI requires both residuals to be nonnegative and complementary. A negative value of R_{obs} means that the candidate post-trade value is lower than the current continuation value and therefore signals an intervention opportunity. A differentiable soft trigger is

$$\alpha_{\theta,\phi}(t, p, \mathbf{w}) = \text{sigmoid}(-\zeta_{\text{tr}} R_{\text{obs}}(t, p, \mathbf{w})), \quad (99)$$

although two alternatives are possible: a separate trade/no-trade score head, or always evaluating (95) and interpreting $\Gamma_\phi \approx \mathbf{w}$ as no trade.

A schematic critic objective is

$$\begin{aligned} \mathcal{L}_{\text{critic}}^{\text{QVI}} = & \mathcal{L}_{\text{terminal}} + \lambda_c \mathbb{E}[(R_{\text{cont}}^-)^2] + \lambda_o \mathbb{E}[(R_{\text{obs}}^-)^2] \\ & + \lambda_{co} \mathbb{E}[(R_{\text{cont}} R_{\text{obs}})^2], \end{aligned} \quad (100)$$

where $r^- = \min(r, 0)$. The actor can combine the simulated terminal objective with feasibility and trigger regularization,

$$\mathcal{L}_{\text{actor}}^{\text{QVI}} = \mathcal{L}_{\text{pathwise}} + \lambda_\Gamma \mathcal{L}_{\text{feas}} + \lambda_{\text{tr}} \mathcal{L}_{\text{trigger}}. \quad (101)$$

In principle, this design allows pure QVI-residual fitting, pure pathwise policy optimization, or a hybrid. A practical future route is to initialize the intervention actor from a trained Direct Boundary RL policy and then introduce the QVI residuals gradually as a curriculum, rather than learning the critic, trigger, and intervention map simultaneously from scratch.

7.4 Why the reported Stage 2 uses Direct Boundary RL

The QVI formulation is the theoretically natural transaction-cost extension, but the preliminary critic-based RL implementation was not stable enough for the reported study. The main numerical difficulties were:

- the state dimension increases through the current weight vector $\mathbf{w}$;
- the minimum in (83) creates a non-smooth continuation/intervention boundary;
- intervention samples are sparse and the trigger can collapse to always-trade or never-trade behavior;

- critic errors enter both the continuation residual and the intervention target $\mathcal{M}_\phi V$;
- the terminal mean–variance signal is delayed, while filtering and transaction costs add further pathwise noise.

In trial runs, the QVI residuals and intervention trigger did not converge reliably and produced unstable actor updates. The final experiment therefore uses *Direct Boundary RL*: a restricted no-trade policy is optimized directly through simulated trajectories. Here “RL” refers to sequential pathwise policy learning from environment interactions; the method does not use a separate transaction-cost critic.

The distinction between the theoretical benchmark and the reported objective is important. The QVI above retains the Stage 1 mean–variance terminal condition, whereas the implemented Direct Boundary RL policy is trained using the terminal utility and regularization objective in (113). Stage 2 is therefore not claimed to solve the exact transaction-cost extension of the Stage 1 problem. The frozen Stage 1 policy is used as a state-dependent structural feature and advanced start, and all reported Stage 2 policy comparisons are interpreted under the separate transaction-cost-aware objective.

7.5 From a frictionless center to an inaction region

Let the implemented Stage 1 center weight be

$$\mathbf{w}_k^* = \mathbf{w}^*(t_k, x_k, p_k) := \frac{\mathbf{u}^*(t_k, x_k, p_k)}{x_k}, \quad x_k > 0. \quad (102)$$

Because the risky-dollar actor is proportional to $x_k - \omega$, this normalized center is generally wealth dependent. Under proportional costs, mechanically rebalancing to $\mathbf{w}_k^*$ at every step can destroy value through turnover. Motivated by the control-limit result of Taksar et al. [13], Stage 2 learns a no-trade region around this wealth-dependent center.

The Stage 2 pre-trade state is

$$o_k = [\tau_k, x_k^-, p_k, \mathbf{w}_k^-, \mathbf{w}_k^*, \mathbf{w}_k^- - \mathbf{w}_k^*], \quad \tau_k = \max(0, 1 - t_k/T). \quad (103)$$

A feedforward network predicts positive lower and upper gaps,

$$\mathbf{g}_k^\ell = \text{softplus}(h_\phi^\ell(o_k)) + \varepsilon, \quad \mathbf{g}_k^u = \text{softplus}(h_\phi^u(o_k)) + \varepsilon, \quad (104)$$

and defines

$$\underline{\mathbf{w}}_k = \mathbf{w}_k^* - \mathbf{g}_k^\ell, \quad \overline{\mathbf{w}}_k = \mathbf{w}_k^* + \mathbf{g}_k^u. \quad (105)$$

First apply the componentwise no-trade-band projection

$$\mathbf{w}_k^{\text{band}} = \min(\max(\mathbf{w}_k^-, \underline{\mathbf{w}}_k), \overline{\mathbf{w}}_k). \quad (106)$$

The reported specification also imposes the hard gross-leverage constraint through

$$\mathbf{w}_k^+ = \Pi_{\mathcal{W}}(\mathbf{w}_k^{\text{band}}), \quad \mathcal{W} = \{\mathbf{w} : \|\mathbf{w}\|_1 \leq L_{\max}\}, \quad L_{\max} = 1.5, \quad (107)$$

where $\Pi_{\mathcal{W}}$ denotes projection onto the admissible leverage set, together with any additional implementation-specific box constraints. Thus no trade occurs inside the band whenever the current position is also feasible; outside the band, the portfolio is moved to the nearest boundary and then made feasible. Unlike a symmetric fixed-width rule, the network permits asymmetric, time-varying, belief-dependent boundaries. The risky-weight vector excludes cash, so the discounted cash weight is $1 - \mathbf{1}^\top \mathbf{w}_k$.

7.6 Wealth recursion and direct objective

To make the trading-time convention explicit, let $(x_k^-, \mathbf{w}_k^-)$ denote pre-trade discounted wealth and risky weights at time t_k . The boundary and feasibility rules in (107) produce the post-trade target $\mathbf{w}_k^+$. With proportional transaction-cost rate κ , define

$$\text{TC}_k = \kappa x_k^- \|\mathbf{w}_k^+ - \mathbf{w}_k^-\|_1, \quad (108)$$

$$x_k^+ = x_k^- - \text{TC}_k, \quad (109)$$

$$\mathbf{u}_k^+ = x_k^+ \mathbf{w}_k^+, \quad (110)$$

$$x_{k+1}^- = x_k^+ + (\mathbf{u}_k^+)^{\top} R_{k+1}^{\text{disc}}, \quad (111)$$

$$\mathbf{u}_{k+1}^- = \mathbf{u}_k^+ \odot (\mathbf{1} + R_{k+1}^{\text{disc}}), \quad (112)$$

where R_{k+1}^{disc} is the vector of discounted simple risky returns over $(t_k, t_{k+1}]$ and $\odot$ denotes componentwise multiplication. The next pre-trade weight is $\mathbf{w}_{k+1}^- = \mathbf{u}_{k+1}^- / x_{k+1}^-$. Elsewhere in the Direct Boundary description, $(x_k, \mathbf{w}_k)$ denotes this pre-trade state $(x_k^-, \mathbf{w}_k^-)$. This notation ensures that transaction costs are charged on the actual change from current pre-trade holdings to the selected post-trade target, rather than on the previous period’s nominal post-trade dollars.

The frozen implementation used for the final tables will follow the same pre-/post-trade timing convention. The final manifest will therefore record the recursion version together with the checkpoint and evaluation seeds. The network is trained by differentiating through (107)–(112). A representative objective is

$$\mathcal{L}_{\text{band}}(\phi) = -\lambda_U \mathbb{E}[U(x_T)] + \lambda_{to} \mathcal{L}_{to} + \lambda_g \mathcal{L}_g + \lambda_{gl} \mathcal{L}_{gl}, \quad (113)$$

where

$$\mathcal{L}_{to} = \mathbb{E}[\|\mathbf{w}_k^+ - \mathbf{w}_k^-\|_1], \quad (114)$$

$$\mathcal{L}_g = \mathbb{E}[\|\mathbf{g}_k^\ell\|_2^2 + \|\mathbf{g}_k^u\|_2^2], \quad (115)$$

$$\mathcal{L}_{gl} = \mathbb{E}[\|\mathbf{w}_k^+\|_1]. \quad (116)$$

The physical proportional transaction cost is already deducted from wealth in (112). The additional turnover term $\mathcal{L}_{to}$ is therefore a deliberate governance and training regularizer—and a simple proxy for unmodeled market impact—rather than a second accounting deduction of the same transaction cost. The gap and gross-leverage terms discourage uninformatively wide bands and positions close to the admissibility boundary.

Typical terminal utilities include logarithmic, square-root, and CRRA utility, all of which require an explicit treatment of nonpositive wealth. An earlier, incompletely archived exploratory run did not record the exact utility, utility floor or bankruptcy penalty, or the coefficients $(\lambda_U, \lambda_{to}, \lambda_g, \lambda_{gl})$, and is not part of the seed-aggregated inference reported starting in Section 8.3. The Phase 1A five-seed replication instead uses logarithmic utility with a fixed risk-aversion parameter, a positivity safeguard applied after transaction-cost deduction, and the loss coefficients recorded in Table 6 and in the archived manifest for every training seed.

Direct Boundary RL is related to the HJB advanced-start strategy of Mulvey et al. [9]: a tractable frictionless solution supplies information that makes the transaction-cost problem easier to learn. The QVI derivation above explains why a no-trade region and center-relative coordinates are natural, but the reported optimizer directly trains the restricted boundary policy class because the attempted QVI critic/intervention actor was unstable. The QVI model is retained as the theoretical benchmark and as a target for future stabilization work.

7.7 Unified training procedure

Algorithm 2 Two-stage structure-guided learning procedure

- 1: Construct return-based regime features and fit a two-state statistical jump model.
 - 2: Estimate $\boldsymbol{\mu}^{(i)}$, $\boldsymbol{\Sigma}^{(i)}$, and λ_i from the inferred labels.
 - 3: Initialize the belief filter, Stage 1 critic, Gaussian actor, and multiplier ω .
 - 4: **for** each Stage 1 training iteration **do**
 - 5: Simulate market paths and update beliefs from observed returns.
 - 6: Sample actions from the structured Gaussian policy and propagate frictionless wealth.
 - 7: Update the critic, actor, and multiplier using the cost-to-go and martingale signals.
 - 8: **end for**
 - 9: Freeze the Stage 1 center map $\mathbf{w}^*(t, x, p)$.
 - 10: **for** each Stage 2 training iteration **do**
 - 11: Simulate common market and belief paths and compute the center path.
 - 12: Predict Direct Boundary RL no-trade gaps and apply the projection rule.
 - 13: Propagate transaction-cost-aware wealth and minimize (113).
 - 14: **end for**
 - 15: Evaluate Stage 1 and Stage 2 policies on common independently generated simulation paths.
-

8 Numerical Study

8.1 Experimental design

The reported experiments use a two-regime, two-risky-asset environment with one trading year, daily time steps, initial discounted wealth $x_0 = 1$, initial belief $p_0 = 0.5$, risk-free rate $r = 0.01$, and Stage 1 target $z = 1.2$. The

common-covariance and regime-dependent covariance Stage 1 models are trained for 9000 iterations. The latter uses the statistical jump model with jump penalty 100, 10 random initializations, and at most 400 iterations. The Phase 1A transaction-cost experiment uses the common-covariance Stage 1 center, estimated-parameter filtering, proportional transaction cost $\kappa = 0.002$, and 1000 common evaluation paths. The leverage cap of 1.5 is applied to the target portfolio weight at each evaluation-time rebalancing decision for every reported Stage 2 comparator; it is an evaluation-side constraint and is unrelated to the Stage 1 actor’s own component-wise action bound (Table 5), which is not enforced during Stage 2 training or evaluation. Under (112), $\kappa = 0.002$ represents 20 basis points per unit of absolute pre-to-post-trade risky-weight turnover, scaled by pre-trade wealth. Reported comparators include daily and monthly center rebalancing, monthly-rebalanced equal weight, and Center+DNNBand.

For Tables 9 and 10, mean shortfall is defined by

$$\text{MSF} := \mathbb{E}[(z - X_T)^+]. \quad (117)$$

Table 5: Selected Stage 1 configuration parameters.

Item	Common covariance	Regime-dependent covariance
Horizon / step	1 year / 1/252	1 year / 1/252
Initial state	$(x_0, p_0) = (1, 0.5)$	$(x_0, p_0) = (1, 0.5)$
Target / entropy	$z = 1.2, \Lambda = 1$	$z = 1.2, \Lambda = 1$
Iterations	9000	9000
Covariance control	common Σ	$\bar{\Sigma}(p)$
Filtering input	estimated parameters	estimated parameters
Regime estimation	common-covariance fit	statistical jump model
Critic steps per outer iteration	1	1
Multiplier EMA parameter	0.5	0.5

Table 6: Stage 2 (Center+DNNBand) configuration for the Phase 1A five-seed replication. All entries are read from the archived `run_config.json`, `eval_config.json`, `aggregation_config.json`, and `run_manifest.csv` for each training seed, not transcribed from memory.

Item	Reported setting
Stage 1 center source	Common-covariance Stage 1 checkpoint, same training seed as the Stage 2 run
Filtering mode	Estimated parameters (<code>filter_mode = estimated.params</code>)
Transaction cost κ	0.002 (proportional, applied to pre-trade wealth)
Evaluation-time leverage cap	1.5, applied to the target weight at each rebalancing decision; not enforced during Stage 2 training and unrelated to the Stage 1 actor’s action bound
Horizon / time step	1 year / 1/252 ($dt = 0.0039683$)
Stage 2 training iterations	800, with 128 simulated episodes per iteration
Training seeds	5 independent seeds (0–4)
Evaluation paths	1000 common held-out paths per training seed, evaluation seed 999
Evaluation scenario set	Identical across all 5 training seeds; confirmed by byte-identical EW-Monthly comparator terminal wealth across seeds
Policy architecture	Feedforward network, hidden width 64, separate softplus lower- and upper-gap heads
Optimizer / learning rate schedule	Adam, initial rate 5×10^{-4} , step decay by 0.5 every 100 iterations
Terminal objective	Log utility ($\gamma_{\text{risk}} = 5$) plus turnover (5×10^{-4}), gap- L_2 (10^{-5}), and gross-leverage (10^{-5}) regularization, as in (113)
Checkpoint policy	Best-validation checkpoint selected every 25 iterations on 64 held-out validation paths; final-iteration checkpoint also retained
Gross/net convention	gross terminal wealth – net terminal wealth – cumulative transaction cost = 0; confirmed to floating-point precision (maximum absolute error $\leq 5.4 \times 10^{-16}$) for every seed and method
Training/evaluation code commit	ba22728 (embedded in every checkpoint); Stage 1 evaluation for seeds 1–2 was re-run under commit 04c4f2e after a path-quoting invocation bug fix, without retraining
Manifest / checkpoint hashes	Recorded per seed in <code>run_manifest.csv</code> (SHA-256)

Tables 7 and 8 distinguish the true simulator from the estimated model used by filtering and control. This separation is essential: learning performance reflects both policy optimization and parameter-estimation error.

Table 7: True and estimated parameters: common-covariance benchmark.

Quantity	True environment	Estimated value used
$\mu^{(1)}$	(0.25, 0.18)	(0.3883, 0.2174)
$\mu^{(2)}$	(−0.73, −0.40)	(−0.4299, −0.3930)
Σ	$\begin{pmatrix} 0.0484 & 0.01188 \\ 0.01188 & 0.0324 \end{pmatrix}$	$\begin{pmatrix} 0.0490 & 0.0117 \\ 0.0117 & 0.0323 \end{pmatrix}$
λ_1	0.36	0.1641
λ_2	2.89	1.7500
Stationary regime-1	0.889	0.914
mass		
Mean holding times	(2.78, 0.35) years	(6.09, 0.57) years

Table 8: True and SJM-estimated parameters: regime-dependent covariance run.

Quantity	True environment	SJM estimate used
$\mu^{(1)}$	(0.25, 0.18)	(0.3830, 0.2759)
$\mu^{(2)}$	(−0.73, −0.40)	(−0.3243, −0.1476)
$\Sigma^{(1)}$	$\begin{pmatrix} 0.0484 & 0.01188 \\ 0.01188 & 0.0324 \end{pmatrix}$	$\begin{pmatrix} 0.0485 & 0.0108 \\ 0.0108 & 0.0322 \end{pmatrix}$
$\Sigma^{(2)}$	$\begin{pmatrix} 0.0625 & 0.0225 \\ 0.0225 & 0.0484 \end{pmatrix}$	$\begin{pmatrix} 0.0574 & 0.0193 \\ 0.0193 & 0.0401 \end{pmatrix}$
λ_1	0.36	0.6801
λ_2	2.89	1.3310
Stationary regime-1	0.889	0.662
mass		
Mean holding times	(2.78, 0.35) years	(1.47, 0.75) years

8.2 Repeated-seed robustness protocol and numerical provenance

The common-covariance Stage 1 and Stage 2 experiments were repeated with five independent training seeds. Each trained policy was evaluated on the same fixed set of 1000 held-out market paths. Reported cross-seed intervals are based on the five seed-level statistics and use the Student- t critical value with four degrees of freedom. These intervals quantify sensitivity to training initialization under the fixed evaluation design; they should not be interpreted as confidence intervals covering all sources of market-model or parameter uncertainty.

Where two methods were evaluated on the same simulated paths, paired differences were first calculated at the episode level, summarized within each training seed, and then aggregated across the five training seeds. The run manifest records the training seed, checkpoint path and hash, evaluation directory, configuration, evaluation-path count, and run status. Two empty Stage 1 evaluation directories produced by a previously identified path-quoting invocation error were excluded; the corrected evaluations used the original completed checkpoints and did not retrain the policies. Because no independent scenario-hash or path-seed record was persisted per path, agreement of the common evaluation set across seeds was instead confirmed by an evaluation-time consistency check: byte-identical deterministic benchmark outputs provided an additional consistency check, since the equal-weight (Stage 1) and equal-weight-monthly (Stage 2) comparators do not depend on the trained policy and are therefore expected to be identical across training seeds if and only if the same underlying evaluation paths were used.

The regime-dependent-covariance experiment reported in Section 8.3.1 was not part of this five-seed replication and remains a single-seed exploratory result (Section 8.3.1).

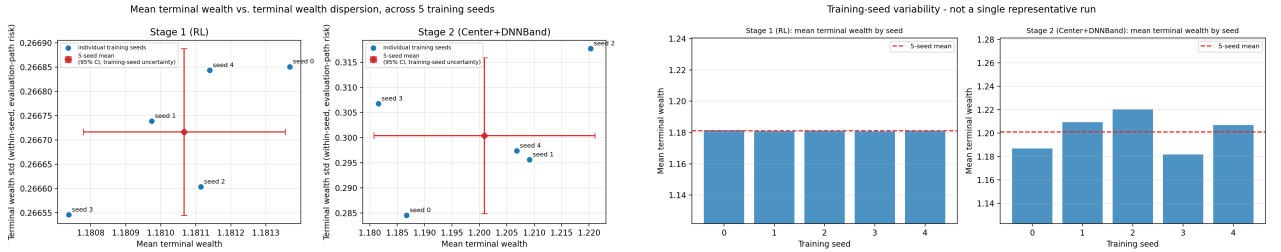
8.3 Stage 1 results

Table 9 summarizes the common-covariance benchmark across five independent training seeds, each evaluated on the same 1000 held-out paths. This case is closest to the theory because both the innovation filter and HJB reduction use common covariance.

Table 9: Common-covariance Stage 1 performance across five independent training seeds. Bracketed values report 95% Student- t intervals across seed-level statistics (four degrees of freedom). Terminal-wealth dispersion (Std. X_T) is calculated across evaluation paths within each seed and is distinct from training-seed uncertainty. Equal weight and minimum variance do not depend on the trained policy and are seed-invariant by construction (zero-width interval).

Method	Mean X_T	Std. X_T (within seed)	$\mathbb{P}(X_T \geq z)$	Mean shortfall	Target MSD
Equal weight	1.0543 (seed-invariant)	0.2688	0.304	0.1943	0.0934
Minimum variance	1.0530 (seed-invariant)	0.2567	0.291	0.1913	0.0874
HJB-guided RL	1.1811 [1.1808, 1.1814]	0.2667	0.566	0.1247	0.0714

Across the five training seeds, HJB-guided RL achieved mean terminal wealth 1.1811 with a 95% training-seed interval of [1.1808, 1.1814], a target-achievement probability of 0.566, and a target mean squared deviation (target MSD) of 0.0714; target MSD is a derived metric computed as $\frac{1}{N_{\text{eval}}} \sum_e (X_{T,e} - z)^2$ from the evaluation terminal-wealth sample, not a native solver output. The between-training-seed standard deviation of mean terminal wealth was approximately 0.000232 – more than three orders of magnitude smaller than the within-seed evaluation-path dispersion (0.2667) – indicating that Stage 1 is extremely stable with respect to training initialization under this fixed evaluation design. Relative to minimum variance, HJB-guided RL raises mean terminal wealth by $1.1811 - 1.0530 = 0.1281$ and raises the target-achievement probability from 0.291 to 0.566; its within-seed standard deviation remains close to the static benchmarks, so the improvement is not explained solely by an increase in terminal dispersion.



(a) Mean terminal wealth vs. within-seed standard deviation, by training seed, with the five-seed mean and its 95% training-seed interval. (b) Mean terminal wealth by individual training seed, Stage 1 (left) and Stage 2 (right).

Figure 2: Training-seed robustness for the common-covariance HJB-guided policy (Stage 1, left panels) and Center+DNNBand (Stage 2, right panels). Both stages are shown together because the same five training seeds and evaluation design underlie both; Stage 1’s near-total overlap of points contrasts with Stage 2’s visibly larger spread.

8.3.1 Exploratory regime-dependent-covariance experiment

The regime-dependent-covariance experiment was not included in the five-seed Phase 1A replication. Its table below reports a separate single-seed exploratory run and is retained only to demonstrate implementability of the approximate filtering-and-control layer; the corresponding figure could not be regenerated from a retained image file and has been removed rather than shown as a placeholder (Section 8.3.1 retains the table and discussion only).

Table 10: Single-seed exploratory regime-dependent-covariance held-out simulation summary. Not part of the five-seed Phase 1A replication; no training-seed interval is reported.

Method	Mean X_T	Std. X_T	Var. X_T	$\mathbb{P}(X_T \geq z)$	Mean shortfall
Equal weight	1.078	0.271	0.073	0.316	0.180
Minimum variance	1.078	0.259	0.067	0.312	0.176
Approximate HJB RL	1.132	0.172	0.030	0.342	0.112

In this single exploratory run, the regime-dependent policy has the highest displayed mean and the lowest dispersion. Its gain in target-hit probability is more modest than in the common-covariance run, and the training curve is noisier. This is consistent with the SJM transition-rate mismatch and with the additional

approximation in (71)–(76). The discrete-time Gaussian filtering–correction step is exact for the adopted observation model, but the belief-averaged control layer is a moment-matched approximation with no verification theorem. These numbers demonstrate promising behavior for the implemented surrogate under a single training initialization; they do not prove optimality for the original regime-dependent continuous-time model, and no multi-seed inference or general dominance claim is made for this experiment.

8.4 Stage 2 results

Table 11 reports the five-seed transaction-cost-aware comparison. The previously reported legacy single-seed illustration (mean net terminal wealth 1.248 for the boundary policy against 500 evaluation paths, versus 1.136/1.147/1.052 for the daily-center, monthly-center, and monthly-equal-weight comparators) used an incompletely archived training seed, checkpoint, and evaluation-path set; it is superseded here by the fully traceable Phase 1A five-seed replication (Table 6) and is not mixed into the table below.

Table 11: Stage 2 transaction-cost-aware performance across five independent training seeds, 1000 common evaluation paths per seed. Bracketed net-terminal-wealth values report 95% Student- t intervals across training seeds. Band width is a structural quantity defined only for Center+DNNBand; “–” denotes not applicable, not zero.

Method	Net X_T	Gross X_T	Cum. tc	Turnover	Trades	Band width	$\mathbb{P}(X_T \geq z)$
Center+DNNBand	1.2009 [1.1807, 1.2211]	1.2196	0.0187	0.0397	175.8	1.740	0.490
CenterOnly, daily	1.1099 [1.1097, 1.1101]	1.1595	0.0496	0.1100	252.0	–	0.426
CenterOnly, monthly	1.1229 [1.1228, 1.1231]	1.1349	0.0120	0.0258	12.0	–	0.437
Equal weight, monthly	1.0372 (seed-invariant)	1.0401	0.0029	0.0059	12.0	–	0.281

Table 12: Selected Stage 2 paired comparisons. Differences are computed at the evaluation-path level within each training seed, averaged within the seed, and then aggregated across five training seeds; 95% intervals use the Student- t critical value with four degrees of freedom. The Stage 2-versus-Stage 1 row is a descriptive comparison across different terminal objectives, not a claim of dominance (Section 8.4). A complete set of eleven paired comparisons, including all pairwise method differences and gross-minus-net decompositions, is reported in Appendix C.

Comparison	Mean difference	95% training-seed interval	Excludes zero?
Center+DNNBand – CenterOnly, daily	0.0910	[0.0709, 0.1112]	yes
Center+DNNBand – CenterOnly, monthly	0.0780	[0.0578, 0.0981]	yes
Center+DNNBand – Equal weight, monthly	0.1637	[0.1436, 0.1839]	yes
Stage 2 net – Stage 1 terminal wealth	0.0198	[−0.0003, 0.0400]	no

Center+DNNBand achieved mean net terminal wealth 1.2009 (95% training-seed interval [1.1807, 1.2211]), mean gross terminal wealth 1.2196, mean cumulative transaction cost 0.0187, mean turnover 0.0397, a mean of 175.8 trades, and mean band width 1.740. Its paired difference against daily center rebalancing was 0.0910 with interval [0.0709, 0.1112], which excludes zero. Center+DNNBand reduced the mean number of trades from 252 under daily center rebalancing to 175.8, a reduction of approximately 30%. This reduction in trade frequency accompanies, but is not shown to causally explain, the net-wealth improvement: separating the contribution of trade frequency from boundary asymmetry, belief conditioning, and time-to-go conditioning would require an ablation study that has not been performed.

The mean descriptive difference between Stage 2 net terminal wealth and Stage 1 terminal wealth was 0.0198, with a 95% training-seed interval of [−0.0003, 0.0400]. Because the interval includes zero and the two stages optimize different terminal objectives (Stage 1 a precommitted mean–variance criterion, Stage 2 a transaction-cost-aware utility objective), this comparison does not establish that Stage 2 dominates Stage 1. It is also notable that, despite its higher mean terminal wealth, Center+DNNBand’s target-achievement probability (0.490) is lower than Stage 1’s (0.566): the two stages differ not only in mean level but in the shape of the terminal-wealth distribution relative to the target z , so policies should not be ranked by mean terminal wealth alone.

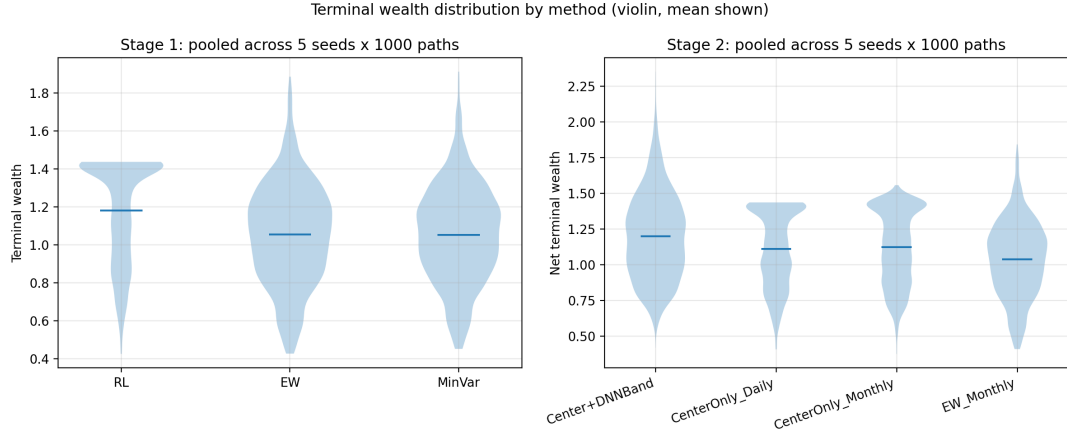


Figure 3: Terminal wealth distributions pooled across five training seeds and 1000 evaluation paths each. Left: Stage 1 (RL, equal weight, minimum variance). Right: Stage 2 (Center+DNNBand and the three comparators), net of transaction costs. Horizontal bars mark the pooled mean.

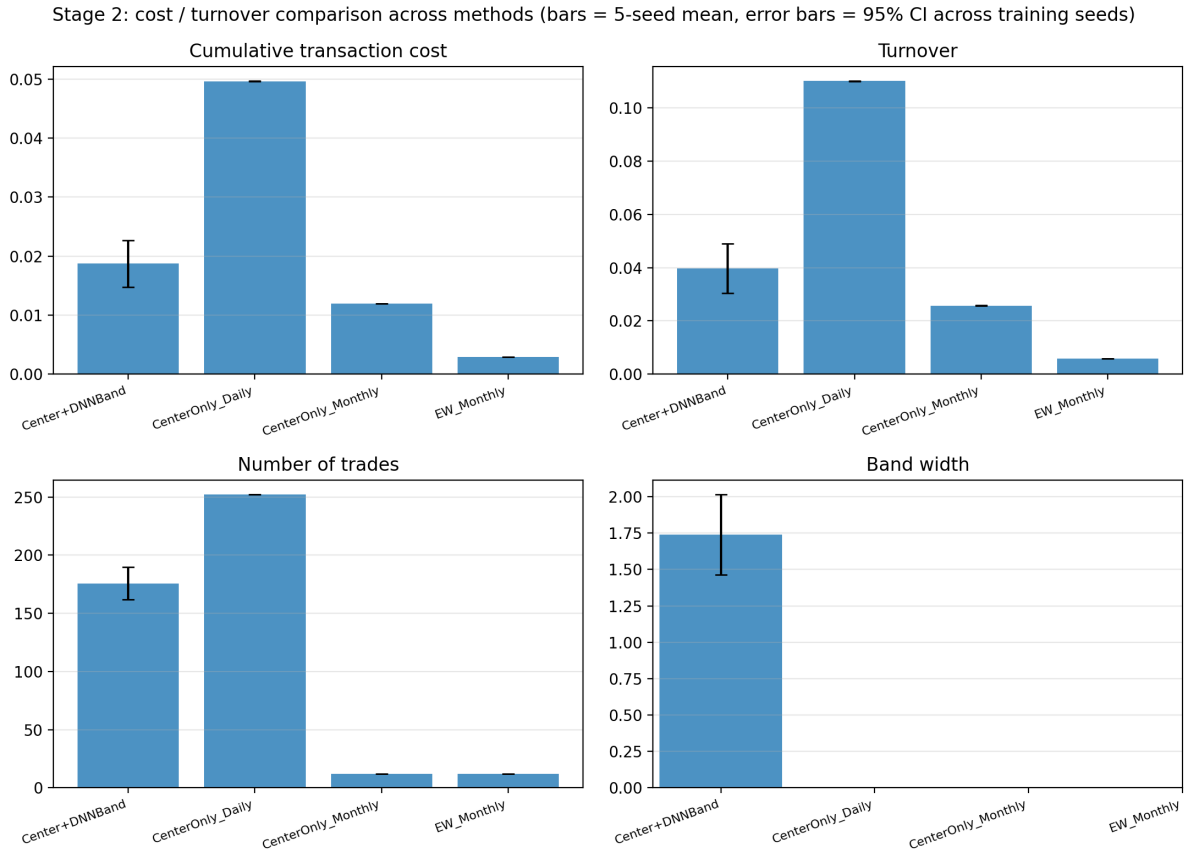


Figure 4: Stage 2 cost and turnover comparison across methods: cumulative transaction cost, turnover, trade count, and band width. Bars show the five-seed mean; error bars show the 95% training-seed interval. Band width bars are blank for methods without a band-width concept (structural, not zero).

The previously reported single-seed Stage 2 training diagnostics (terminal-objective and direct-boundary-loss learning curves) could not be regenerated from a retained image file for this revision and have been removed rather than shown as placeholders; the underlying per-iteration training metrics remain available in the archived run outputs for each of the five training seeds.

9 Discussion

9.1 Practical implications

The results illustrate how an institutional investment rule can separate three economically distinct tasks: estimating the market environment, selecting a frictionless risk exposure, and deciding whether the benefit of rebalancing justifies its cost. The belief state is not merely a forecasting output; it enters the actor, critic, and transaction-cost policy as a decision variable. Likewise, the no-trade region provides an interpretable governance rule: trade only when the current position departs sufficiently from the belief- and horizon-dependent center.

This decomposition also clarifies why reinforcement learning is useful in this setting. The action is naturally a portfolio allocation, the state representation can be constructed to summarize economically relevant information, and the policy must account for the effect of current trading on future wealth and transaction opportunities. Incorporating domain knowledge into the state and policy does more than reduce dimension: the belief has a probabilistic interpretation, the Gaussian Stage 1 actor separates myopic and hedging demand, and the Stage 2 boundary explains when rebalancing is deferred. The same framework permits domain knowledge to enter the objective through target wealth, entropy, turnover, leverage, and transaction-cost terms.

For insurers and pension funds, the target wealth can serve as a reduced-form accumulation or funding objective, while the trading band can reflect an operational preference for stable allocations and controlled turnover. Across the five-seed Phase 1A replication, the no-trade region reduced the mean number of trades by approximately 30% relative to daily center rebalancing (Section 8.4), which can be read as evidence that the framework offers an operational turnover-discipline benefit in addition to its net-performance statement, not only as a net-wealth result. The current results are methodological demonstrations rather than investment recommendations, and actual market impact, transaction-cost nonlinearity, and liability dynamics are not modeled. Their practical value lies in providing a transparent architecture that could later accommodate liability cash flows, capital constraints, prescribed asset limits, and institution-specific transaction budgets.

9.2 What is exact and what is approximate

The common-covariance Stage 1 model has the strongest theoretical status: filtering produces an observable diffusion, the entropy-regularized HJB yields the Gaussian optimizer, and the wealth-quadratic ansatz produces exact lower-dimensional coefficient PDEs under the stated regularity conditions. The SJM layer is an estimation methodology that supplies parameters to this filter; it is not part of the control verification theorem.

For regime-dependent covariance, the discrete-time Gaussian prediction-correction filter is exact for the specified hidden Markov observation model, but the belief-averaged effective diffusion and its HJB are moment-matched approximations. For transaction costs, the QVI is the theoretical benchmark and the center-relative critic/intervention actor is exploratory. Center+DNNBand, the reported Direct Boundary RL method, produced a numerically usable pathwise policy across all five training seeds, with a separate terminal objective; it is not a numerical solution of that QVI.

9.3 Lessons from theory-guided learning

The experiments suggest an asymmetric role for structural theory. In Stage 1, the common-covariance HJB identifies the Gaussian policy family, the linear dependence of its mean on wealth, the quadratic critic, and the lower-dimensional coefficient functions. These restrictions remove large parts of the learning problem without materially narrowing the theorem-level optimizer. The resulting actor-critic provided usable training and favorable displayed results under both common and regime-dependent covariance. The very small between-seed dispersion in Stage 1 (between-training-seed standard deviation of mean terminal wealth ≈ 0.000232 , against a within-seed evaluation-path standard deviation of 0.267) is consistent with the stabilizing role of the HJB-restricted actor and critic, although the current experiments do not causally identify that restriction as the source of stability: because a controlled structured-versus-unstructured ablation has not yet been completed, the present evidence does not identify the restrictions themselves as the sole cause of the observed performance.

The Stage 2 experience is different. The QVI identifies the enlarged state, the intervention-versus-continuation logic, and the relevance of a no-trade region, but it does not determine the finite-horizon multidimensional boundary geometry as sharply as the Stage 1 HJB determines the Gaussian actor. Jointly learning the QVI critic, obstacle residual, trigger, and intervention map was unstable, while preliminary generic deep-RL and Transformer-style formulations also failed to produce a sufficiently reliable policy within the available experiments. This comparison does not identify the encoder architecture or the absence of a particular parametrization as the unique cause of instability. The Direct Boundary architecture instead imposed economically meaningful

center-relative no-trade geometry, retained feasibility constraints and pathwise transaction costs, and produced a numerically usable implementation. Consistent with a less sharply identified boundary geometry, Stage 2’s between-training-seed standard deviation of mean net terminal wealth was approximately 0.0163 – about 70 times larger than Stage 1’s between-training-seed standard deviation of mean terminal wealth (≈ 0.000232) – indicating materially greater sensitivity to training initialization than Stage 1 under the same five-seed protocol.

The methodological lesson is not that stronger structural restrictions are always preferable. Theory-guided parametrization is most effective when the underlying control equation identifies the relevant functional form with sufficient precision. When theory supplies only local or qualitative information about a free boundary, a softer inductive bias combined with simulation-based learning may be preferable to enforcing a misspecified low-dimensional geometry. The formula-based QVI actor remains valuable as an interpretable target and potential regularizer, but not yet as the reported optimizer.

9.4 Sources of performance and model risk

Performance in the regime-aware pipeline can be affected at five distinct layers:

- (1) **Regime identification error:** the SJM may assign observations to economically imperfect states.
- (2) **Parameter error:** regime-specific drifts, covariances, and transition intensities may be estimated inaccurately even conditional on the labels.
- (3) **Filtering error:** misspecified parameters distort the posterior belief supplied to the controller.
- (4) **Control approximation error:** the belief-averaged regime-dependent covariance model may differ from the original partially observed dynamics.
- (5) **Optimization error:** actor-critic or boundary training may converge slowly, become unstable, or settle at a poor local solution.

The current regime-dependent experiment combines all five effects, so its noisier learning curve cannot be attributed to a single source. The SJM transition-rate mismatch provides direct evidence of parameter risk, while the additional f_p^2/f term and moment matching identify a separate control-model risk. The displayed approximate HJB policy has higher mean terminal wealth and lower dispersion than the static benchmarks, which supports the usefulness of the structural approximation in this setting, but it does not establish universal dominance or isolate the source of the improvement.

The SJM produced a practically usable persistent segmentation in the reported experiment, but the present study does not establish superiority over HMM-based estimation. A controlled comparison should examine label stability, parameter error, posterior filtering accuracy, and downstream portfolio performance under common data and evaluation paths. The current claim is limited to the operational value of combining transparent offline SJM segmentation with online Bayesian belief updating.

Because the reported SJM features are constructed primarily from an equal-weight return index, covariance regimes are identified only indirectly through univariate aggregate-return features. Incorporating cross-sectional dispersion, realized correlation, or rolling covariance features is an important robustness extension, especially because covariance-state variation is a central motivation for the regime-dependent model.

The Stage 2 results have a similarly specific interpretation. Daily rebalancing to the frictionless center has lower mean net wealth than monthly rebalancing, consistent with repeated proportional costs offsetting the marginal tracking benefit. Center+DNNBand has the highest mean net wealth among the reported comparators and reduces mean trade count relative to daily rebalancing, and turnover, cumulative transaction cost, trade count, and band width are now reported as separate summary statistics (Table 11, Figure 4) rather than inferred only from net wealth. However, the current evidence does not determine whether the net-wealth difference comes primarily from lower trading frequency, asymmetric boundaries, belief conditioning, or time-to-go conditioning; separating these mechanisms causally requires an ablation analysis that has not been performed.

9.5 Limitations and robustness status

The common-covariance Stage 1 and Stage 2 experiments have now been verified across five independent training seeds on a common evaluation design (Section 8.3–8.4), which resolves the single-seed limitation noted in earlier drafts for those two components specifically. The following limitations remain:

- (1) Five training seeds are informative about sensitivity to training initialization but remain a small sample; the reported Student- t intervals should not be read as tight population-level confidence statements.

- (2) The regime-dependent-covariance experiment (Section 8.3.1) remains single-seed and exploratory; it is not part of the five-seed replication.
- (3) Transaction-cost sensitivity (varying κ and the leverage cap) has not been examined.
- (4) An actor-signal ablation separating the martingale and full-horizon terms, and a structured-versus-unstructured policy-class ablation, have not been performed.
- (5) Per-timestep wealth trajectories were not persisted by the evaluation code, so a time-resolved (rather than terminal-only) multi-seed wealth-path analysis is not currently available.
- (6) A true, time-resolved cap-saturation rate has not been measured. The reported `share_paths_avg_leverage_near_cap` diagnostic (threshold: whole-path average gross leverage ≥ 1.980) is a path-average proxy over the fraction of evaluation paths, not a per-timestep saturation rate, and should not be read as one.
- (7) Separate lower- and upper-boundary summary statistics were not persisted by the evaluation code; only the combined band width is reported.
- (8) Stage 1 and Stage 2 optimize different terminal objectives, so comparisons between them (Section 8.4) are descriptive, not a test of dominance under a common objective.
- (9) Filtering quality depends on the regime segmentation and parameter estimates, and the estimated transition persistence differs materially from the simulator in the regime-dependent-covariance experiment; the associated control layer is moment-matched rather than exact, and $\tilde{c}(p)$ has not been identified from a continuous-time filtering theorem.
- (10) The attempted QVI-based RL was unstable, and hard componentwise projection in Center+DNNBand is interpretable but does not represent the most general multidimensional intervention geometry.
- (11) Liabilities, solvency capital, taxes, market impact, and governance constraints are not explicitly modeled.

9.6 Future research

The transaction-cost QVI derived in Section 7 provides the theoretical target,

$$\min\{-V_t - \mathcal{L}V, \mathcal{M}V - V\} = 0, \quad (118)$$

where $\mathcal{M}$ accounts for proportional wealth loss and a post-trade target map. The next methodological step is not to formulate this QVI again, but to stabilize its numerical learning: continuation/intervention sampling, obstacle-compatible losses, target networks for $\mathcal{M}V$, curriculum learning from Direct Boundary RL, and smooth trigger approximations are natural candidates.

Broader sensitivity to transaction-cost rates, entropy weights, and filtering misspecification, together with ablations separating the effects of the belief state, HJB-guided parametrization, martingale signal, full-horizon residual, symmetric versus asymmetric bands, and QVI-derived priors, remain future work unless completed before submission. Stronger benchmarks should include a full-information regime oracle retrained under that information structure, a true-parameter filtered policy, a myopic policy without the belief-hedging term, and a fixed symmetric band. Supplying the true regime directly to a policy trained only on filtered beliefs would instead be an out-of-distribution stress test, not a formal oracle comparison. An actuarial extension should introduce stochastic liability cash flows, funding-ratio or surplus objectives, regulatory capital constraints, and asset eligibility limits so that the portfolio controller becomes a component of a complete ALM problem.

10 Conclusion

This paper starts from the view that reinforcement learning is well matched to dynamic portfolio choice when the economically relevant state and admissible policy class are constructed carefully. The latent regime is converted into a filtered belief state, the frictionless actor is restricted by the HJB, and transaction costs are handled through a second-stage no-trade policy around the learned center. The resulting architecture combines state estimation, sequential control, and rebalancing discipline without treating portfolio selection as an unrestricted state-to-weight prediction problem.

The main contributions are the operational multi-asset implementation of partially observed mean-variance actor-critic learning; the integration of SJM-based regime estimation with online filtering; a discretized Gaussian filtering and approximate HJB extension for regime-dependent covariance; and a two-stage transaction-cost framework that connects a QVI benchmark, a formula-based intervention actor, and Direct Boundary learning.

The covariance extension is important because changing volatility and correlation directly affect diversification and risk control, even when expected-return differences are difficult to estimate reliably.

The five-seed common-covariance study provides different robustness conclusions for the two learning stages. Stage 1 converged to nearly identical held-out performance across training seeds, with mean terminal wealth 1.1811 and very small between-seed dispersion. In Stage 2, Center+DNNBand achieved the highest average net terminal wealth among the reported transaction-cost-aware comparators and retained a positive paired difference relative to daily and monthly center rebalancing. It also reduced mean trade count by approximately 30% relative to daily rebalancing. However, Stage 2 was materially more sensitive to training initialization, and its descriptive net-wealth difference from Stage 1 had a 95% interval that included zero. The evidence therefore supports robustness relative to the reported Stage 2 comparators, not a claim that Stage 2 statistically dominates the frictionless Stage 1 policy.

The regime-dependent-covariance evidence remains exploratory and single-seed (Section 8.3.1) and is not covered by the five-seed robustness statements above. Center+DNNBand is not a numerical solution of the transaction-cost QVI derived in this paper; it is a structurally motivated but separately optimized implementation. The paper’s contribution is the selective use of financial structure – filtering, HJB-consistent policy geometry, and a frictionless center – to constrain what must be learned, not a claim that either stage is a verified optimal control solution. An actuarial application of this framework would require a further extension to a complete asset–liability management setting, incorporating stochastic liabilities, capital constraints, and institution-specific admissibility rules.

The broader methodological conclusion is that the contribution lies not in deep learning alone, but in selectively using financial structure to determine the information state, restrict policy geometry, and design the objective, while retaining flexible learning where the control theory does not identify the optimal functional form sufficiently sharply. Stage 1 uses strong low-dimensional restrictions because the Gaussian actor and quadratic critic are characterized by the HJB. In Stage 2, the free-boundary geometry is less completely identified, so flexible simulation-based boundary learning provided the usable implementation, while the QVI and center-relative formulas remain theoretical and architectural guides. For actuarial and institutional applications, this combination of interpretable belief states, structured policies, and transaction-aware rebalancing offers a tractable portfolio-control building block. A complete ALM implementation would additionally require stochastic liabilities, capital constraints, and institution-specific admissibility rules.

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A Explicit QVI Continuation Coefficients

For compactness, suppress the arguments $(p, \mathbf{w})$ and define

$$s = \|\bar{\sigma}\|^2, \quad q = \|\beta\|^2, \quad R = DD^\top, \quad c_p = \bar{\sigma}^\top \beta, \quad c_w = D\bar{\sigma}, \quad c_{pw} = D\beta. \quad (119)$$

The generator associated with (81) is

$$\begin{aligned} \mathcal{L}V = & x\bar{\mu}V_x + \mu_p V_p + b_w^\top \nabla_w V + \frac{1}{2}x^2 s V_{xx} + \frac{1}{2}q V_{pp} + \frac{1}{2}\text{tr}(R\nabla_{ww}^2 V) \\ & + x c_p V_{xp} + x c_w^\top \nabla_w V_x + c_{pw}^\top \nabla_w V_p. \end{aligned} \quad (120)$$

Substitution of (84) into (86) gives

$$\begin{aligned} \Phi_2 = & -A_t - 2\bar{\mu}A - sA - \mu_p A_p - b_w^\top \nabla_w A - \frac{1}{2}q A_{pp} - \frac{1}{2}\text{tr}(R\nabla_{ww}^2 A) \\ & - 2c_p A_p - 2c_w^\top \nabla_w A - c_{pw}^\top \nabla_w A_p, \end{aligned} \quad (121)$$

$$\begin{aligned} \Phi_1 = & -B_t - \bar{\mu}(B + 2\omega A) - 2\omega s A - \mu_p B_p - b_w^\top \nabla_w B - \frac{1}{2}q B_{pp} - \frac{1}{2}\text{tr}(R\nabla_{ww}^2 B) \\ & - c_p(B_p + 2\omega A_p) - c_w^\top (\nabla_w B + 2\omega \nabla_w A) - c_{pw}^\top \nabla_w B_p, \end{aligned} \quad (122)$$

$$\begin{aligned} \Phi_0 = & -C_t - \omega \bar{\mu}B - \omega^2 s A - \mu_p C_p - b_w^\top \nabla_w C - \frac{1}{2}q C_{pp} - \frac{1}{2}\text{tr}(R\nabla_{ww}^2 C) \\ & - \omega c_p B_p - \omega c_w^\top \nabla_w B - c_{pw}^\top \nabla_w C_p. \end{aligned} \quad (123)$$

Equations (121)–(123) show why a third coefficient B is required: even if $B(T, p, \mathbf{w}) = 0$, the terms proportional to ωA act as sources for the linear equation away from maturity.

B Center-Relative QVI Critic and Intervention Actor

Let $\tau = T - t$, $\delta = \mathbf{w} - \mathbf{w}^{\text{ref}}(t, p)$, and

$$\|\delta\|_{M(p)}^2 = \delta^\top M(p) \delta, \quad (124)$$

where $M(p)$ may be I , $\bar{\Sigma}(p)^{-1}$, or a diagonal inverse-variance metric. A low-dimensional critic consistent with positivity of A and the parity of the coefficient equations is

$$A_\theta(\tau, p, \delta) = \exp \left(a_0(\tau, p) + a_2(\tau, p) \|\delta\|_{M(p)}^2 + a_4(\tau, p) \|\delta\|_{M(p)}^4 \right), \quad (125)$$

$$B_\theta(\tau, p, \delta) = b_1(\tau, p)^\top \delta + b_3(\tau, p)^\top \delta \|\delta\|_{M(p)}^2, \quad (126)$$

$$C_\theta(\tau, p, \delta) = c_0(\tau, p) + c_2(\tau, p) \|\delta\|_{M(p)}^2 + c_4(\tau, p) \|\delta\|_{M(p)}^4. \quad (127)$$

Each coefficient surface in (τ, p) can be represented by a low-order polynomial, as in Stage 1. The intervention actor may be written

$$\Gamma_\phi(t, p, \mathbf{w}) = \mathbf{w} + G_\phi(\tau, p, \delta) = \mathbf{w}^{\text{ref}}(t, p) + \delta + G_\phi(\tau, p, \delta), \quad (128)$$

with a radial-directional model

$$G_\phi = -s_\phi(\tau, p, \delta) \delta + R_\phi(\tau, p, \delta), \quad s_\phi = \text{sigmoid} \left(\eta_0(\tau, p) + \eta_2(\tau, p) \|\delta\|_{M(p)}^2 + \eta_4(\tau, p) \|\delta\|_{M(p)}^4 \right). \quad (129)$$

The radial first model sets $R_\phi \equiv 0$, so $s_\phi = 0$ gives no trade and $s_\phi = 1$ moves directly to the normalized reference center. This ensures $G_\phi(\tau, p, 0) = 0$ and biases interventions toward that center. A feasibility layer can enforce simplex, box, or leverage constraints. Because the implemented center (102) depends on wealth, this appendix basis is an architectural approximation and does not claim exact closure after the wealth-dependent change of coordinates.

For the cost-minimization convention, define

$$R_{\text{cont}} = -V_{\theta, t} - \mathcal{L}V_\theta, \quad R_{\text{obs}} = \mathcal{M}_\phi V_\theta - V_\theta. \quad (130)$$

The exact QVI requires

$$R_{\text{cont}} \geq 0, \quad R_{\text{obs}} \geq 0, \quad R_{\text{cont}} R_{\text{obs}} = 0. \quad (131)$$

A soft residual objective may combine terminal supervision with penalties for negative continuation/obstacle residuals and complementarity:

$$\mathcal{L}_{\text{QVI}} = \mathcal{L}_{\text{terminal}} + \lambda_c \mathbb{E}[(R_{\text{cont}}^-)^2] + \lambda_o \mathbb{E}[(R_{\text{obs}}^-)^2] + \lambda_{co} \mathbb{E}[(R_{\text{cont}} R_{\text{obs}})^2], \quad (132)$$

where $r^- = \min(r, 0)$. This was the intended critic-based Stage 2 direction. In practice, simultaneous fitting of these residuals and the intervention actor was unstable, motivating the Direct Boundary RL implementation reported in the main text.

C Complete Phase 1A Paired Comparisons

Table 13 reports all eleven Stage 2 paired comparisons computed for the five-seed Phase 1A replication, extending the four comparisons selected for the main text (Table 12). Methodology is identical: differences are computed at the evaluation-path level within each of the five training seeds, averaged within the seed, and then aggregated across seeds using the Student- t critical value with four degrees of freedom. Rows marked “descriptive” compare methods with different rebalancing rules or, in the Stage 2-versus-Stage 1 case, different terminal objectives; they are not dominance claims.

Table 13: Complete set of Phase 1A paired comparisons, 5 training seeds $\times$ 1000 evaluation paths per seed.

Comparison	Mean diff.	95% interval	Excl. zero?	Type
Stage 2 net – Stage 1 terminal wealth	0.0198	[−0.0003, 0.0400]	no	descriptive
Gross – net, Center+DNNBand	0.0187	[0.0148, 0.0227]	yes	cost decomposition
Gross – net, CenterOnly daily	0.0496	[0.0496, 0.0497]	yes	cost decomposition
Gross – net, CenterOnly monthly	0.0120	[0.0120, 0.0120]	yes	cost decomposition
Gross – net, EW monthly	0.0029	[0.0029, 0.0029]	yes	cost decomposition
Center+DNNBand – CenterOnly daily	0.0910	[0.0709, 0.1112]	yes	descriptive
Center+DNNBand – CenterOnly monthly	0.0780	[0.0578, 0.0981]	yes	descriptive
Center+DNNBand – EW monthly	0.1637	[0.1436, 0.1839]	yes	descriptive
CenterOnly daily – CenterOnly monthly	−0.0131	[−0.0131, −0.0130]	yes	descriptive
CenterOnly daily – EW monthly	0.0727	[0.0725, 0.0729]	yes	descriptive
CenterOnly monthly – EW monthly	0.0858	[0.0856, 0.0860]	yes	descriptive

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